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Mysore
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VTU Question Papers

CIVIL ENGINEERING

Even Semester

2021-22

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

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18MAT31

Third Semester B.E. Degree Examination, July/August 2022 Transform Calculus, Fourier Series and Numerical Techniques

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Find the Laplace transform,
 (i) $e^{-2t}(2\cos 5t - \sin 5t)$ (ii) $\cosh^2 3t$ (06 Marks)
- b. Find the Laplace transform of the full wave rectifier $f(t) = E \sin \omega t$ $0 < t < \frac{\pi}{\omega}$ having a period $\frac{\pi}{\omega}$. (07 Marks)
- c. Find the inverse Laplace transform $\left[\frac{s^2 + 4}{s(s+4)(s-4)} \right]$. (07 Marks)

OR

- 2 a. Find the Laplace transform, $\frac{\cos at - \cos bt}{t}$. (06 Marks)
- b. Solve by using Laplace transform method $y''(t) + 2y'(t) - y'(t) - 2y(t) = 0$, given $y(0) = y'(0) = 0$ and $y'(0) = 6$ (07 Marks)
- c. Express the function $f(t)$ in terms of unit step function and hence find its inverse LT,

$$f(t) = \begin{cases} \cos t & 0 < t \leq \pi \\ 1 & \pi < t \leq 2\pi \\ \sin t & t > 2\pi \end{cases}$$
 (07 Marks)

Module-2

- 3 a. Obtain the Fourier series of $f(x) = \frac{\pi - x}{2}$, in $0 < x < 2\pi$. Hence deduce that
 $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$. (06 Marks)
- b. Show that the sine half range series for the function, $f(x) = Lx - x^2$, in $0 < x < L$ is

$$\frac{8L^2}{\pi^3} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^3} \sin \left(\frac{(2n+1)\pi x}{L} \right)$$
 (07 Marks)
- c. Obtain the Fourier series of y up to the first harmonics for the following values :

x°	45	90	135	180	225	270	315	360
y	4.0	3.8	2.4	2.0	-1.5	0	2.6	3.4

(07 Marks)

OR

- 4 a. Expand the function $f(x) = x \sin x$, as a Fourier series in the interval $-\pi \leq x \leq \pi$. Deduce that $\frac{1}{1,3} - \frac{1}{3,5} + \frac{1}{5,7} - \dots = \frac{\pi-2}{4}$ (06 Marks)

- b. Obtain the half range cosine series of $f(x) = x \sin x$ $0 \leq x \leq \pi$. (07 Marks)

- c. Obtain the constant term and the first three coefficients in the Fourier cosine series for y using the following data :

x	0	1	2	3	4	5
y	4	8	15	7	6	2

(07 Marks)

Module-3

- 5 a. Find the complex Fourier transform of the function, $f(x) = \begin{cases} 1 & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases}$.

Hence evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$. (06 Marks)

- b. If $f(z) = \frac{2z^2 + 3z + 12}{(z-1)^4}$ find the value of u_0, u_1, u_2, u_3 (07 Marks)

- c. Solve by using z-transforms, $u_{n+2} + 5u_{n+1} + 6u_n = 2^n$; $u_1 = 0, u_0 = 0$ (07 Marks)

OR

- 6 a. Find the Fourier sine transform of e^{-ax} , $a > 0$. (06 Marks)

- b. Find the Fourier sine and cosine transform of $2e^{-3x} + 3e^{-2x}$. (07 Marks)

- c. Solve by using Z-transforms, $y_{n+2} + 2y_{n+1} + y_n = n$, with $y(0) = 0 = y$ (07 Marks)

Module-4

- 7 a. Use Taylor's series method to find $y(4.1)$ given that $\frac{dy}{dx} = \frac{1}{x^2 + y}$ and $y(4) = 4$. (06 Marks)

- b. Use Fourth order Runge-Kutta method to solve $(x+y) \frac{dy}{dx} = 1$, $y(0.4) = 1$ at $x = 0.5$. Correct to four decimal places. (07 Marks)

- c. The following table gives the solution of $5xy^1 + y^2 - 2 = 0$, find the value of y at $x = 4.5$ using Milne's Predictor and Corrector formulae, use the corrector formulae twice.

x	4	4.1	4.2	4.3	4.4
y	1	1.0049	1.0097	1.0143	1.0187

(07 Marks)

OR

- 8 a. Using modified Euler's method find y at $x = 0.2$ given $\frac{dy}{dx} = 3x + \frac{y}{2}$, with $y(0) = 1$ taking $h = 0.1$. (06 Marks)

- b. Using Runge-Kutta method of fourth order find $y(0.2)$ for the equation $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$ taking $h = 0.2$ (07 Marks)

- c. Apply Adams-Bashforth method to solve the equation $(y^2 + 1)dy - x^2 dx = 0$, at $x = 1$, given $y(0) = 1, y(0.25) = 1.0026, y(0.5) = 1.0206, y(0.75) = 1.0679$. Apply the corrector formulae twice. (07 Marks)

Module-5

- 9 a. Given $\frac{d^2y}{dx^2} - x_2 \frac{dy}{dx} - 2xy = 1$, $y(0) = \frac{1}{2}$, $y'(0) = 0$, Evaluate $y(0.1)$ using Runge-Kutta method of order 4. (06 Marks)
- b. A necessary condition for the integral $I = \int_{x_1}^{x_2} f(x, y, y') dx$ where $y(x_1) = y_1$ and $y(x_2) = y_2$ to be extremum that $\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$. (07 Marks)
- c. Show that the extremal of the functional $\int_0^1 y^2 \{ 3x(y'^2 - 1) + yy'^3 \} dx$, subject to the conditions $y(0) = 0$, $y(1) = 2$, is the circle $x^2 + y^2 - 5x = 0$. (07 Marks)

OR

- 10 a. Apply Milne's method to compute $y(0.8)$. Given that $\frac{d^2y}{dx^2} = 1 - 2y \frac{dy}{dx}$ and the following table of initial values. (06 Marks)
- | | | | | |
|----|---|--------|--------|--------|
| x | 0 | 0.2 | 0.4 | 0.6 |
| y | 0 | 0.02 | 0.0795 | 0.1762 |
| y' | 0 | 0.1996 | 0.3937 | 0.5689 |
- b. Find the extremal of the functional $\int_a^b (x^2 y'^2 + 2y^2 + 2xy) dx$. (07 Marks)
- c. Prove that Geodesics on a plane are straight line. (07 Marks)

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18CV32

Third Semester B.E. Degree Examination, July/August 2022 Strength of Materials

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain: (i) Stress (ii) Strain (iii) Modulus of Elasticity (06 Marks)
 b. Define four elastic constraints. (08 Marks)
 c. Determine the stresses in various segments of circular bar shown in the Fig.Q1(c). Also compute the total elongation taking Young's modulus $E = 195 \text{ GPa}$.

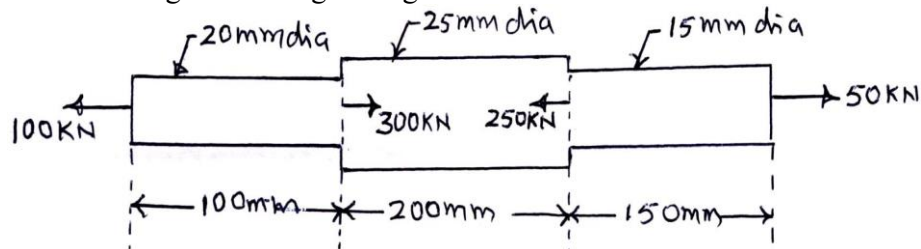


Fig.Q1(c)

(06 Marks)

OR

- 2 a. Derive an expression for the deformation of a rectangular tapering bar of uniform thickness subjected to an axial force. (10 Marks)
 b. A steel tube of 80 mm outer diameter and 10 mm thick is filled into a copper tube of 30 mm inner diameter and 10 mm thick. They are connected rigidly at the ends and subjected to an axial compression of 300 kN load. If length of bar is 600 mm, find the stresses in each material. Take $E_s = 200 \text{ GPa}$ and $E_{cu} = 120 \text{ GPa}$. (05 Marks)
 c. A steel rod of 3 cm diameter and 5 m long is connected to two grips and the rod is maintained at temperature of 95°C . Determine the stresses and pull exerted when temperature falls below 30°C if:
 (i) The ends are not yielding
 (ii) The ends yields by 0.12 cm
 Take $E = 2 \times 10^5 \text{ MN/m}^2$ and $\alpha = 12 \times 10^{-6}/^\circ\text{C}$. (05 Marks)

Module-2

- 3 a. Explain maximum shear stress theory of failure. (08 Marks)
 b. A cast iron pipe 200 mm internal diameter and 50 mm metal thickness carries water under a pressure of 5 N/mm^2 . Calculate the maximum and minimum intensities of circumferential stresses and sketch the distribution of circumferential stress intensity and intensity of radial pressure across the cross section. (12 Marks)

OR

- 4 a. Differentiate between thick and thin cylinders. (02 Marks)
 b. For thin cylinder, derive the equations for circumferential stress and longitudinal stress. (08 Marks)

- c. At a certain point in a strained material the stress conditions shown in Fig.Q4(c). Determine:
- The normal and shear stresses on the inclined plane AB.
 - Principal stresses and principal planes.
 - Maximum shear stresses and their planes.

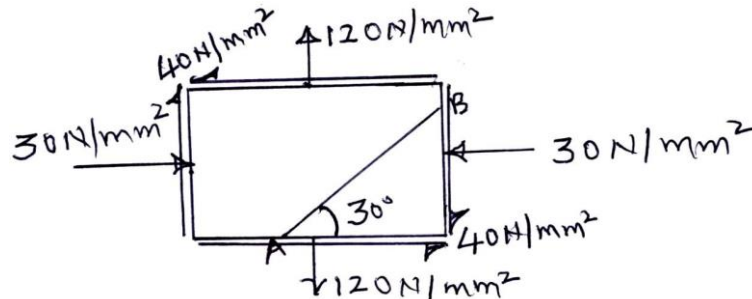


Fig.Q4(c)

(10 Marks)

Module-3

- 5 a. Explain different types of beams and supports. (06 Marks)
- b. Define: (i) Shear force (ii) Bending moment (iii) Point of contra-flexure (06 Marks)
- c. For the cantilever beam shown in Fig.Q5(c), draw shear force and bending moment diagram.

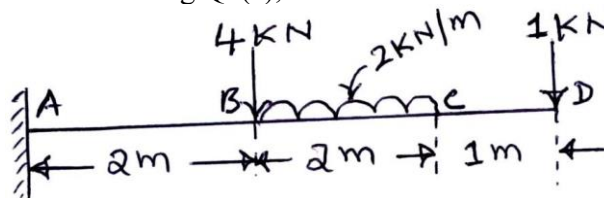


Fig.Q5(c)

(08 Marks)

OR

- 6 a. Derive the relationship between shear force, bending moment and load intensity. (08 Marks)
- b. Draw the shearforce and bending moment diagram indicating principal values for an overhanging beam shown in Fig.Q6(b). Locate point of contraflexure if any.

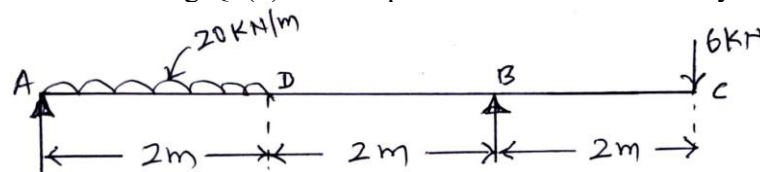


Fig.Q6(b)

(12 Marks)

Module-4

- 7 a. Derive bending equation $\frac{M}{I} = \frac{f}{y} = \frac{E}{R}$ with usual notations. (10 Marks)
- b. A hollow shaft has to transmit 600 KW power at 80 rpm. Torque developed may exceed the mean torque by 40%. Design the suitable section if the working stress is 90 MPa. Take diameter ratio 0.8. What will be the angular twist measured over a length of 2 m? Take $C = 84$ GPa. (10 Marks)

OR

- 8 a. Derive torsion equation $\frac{T}{J} = \frac{C\theta}{l} = \frac{q}{R}$ for a circular shaft subjected to pure torsion. (10 Marks)
- b. Explain pure bending. (04 Marks)

- c. Draw shear stress distribution for an I-shaped section of beam shown in Fig.Q8(c). Shear force on the section is 200 kN.

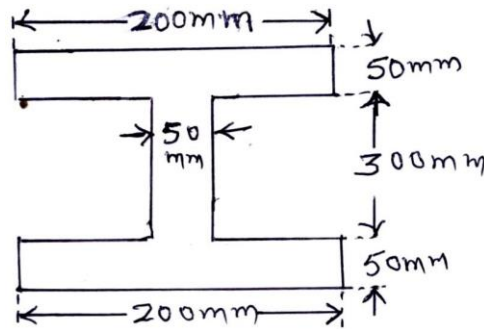


Fig.Q8(c)

(06 Marks)

Module-5

- 9 a. Derive the deflection equation for a beam in the standard form $EI \frac{d^2y}{dx^2} = -M$. (08 Marks)
- b. Define: (i) Slope (ii) Deflection (iii) Elastic curve (06 Marks)
- c. Find the Euler's crippling load for a hollow cylindrical steel column 40 mm external diameter and 4 mm thick. Consider the length of column 2.3 m and hinged at its both the ends. Also determine the crippling load by Rankines formula using constants 335 N/mm^2 and $a = \frac{1}{75000}$. Take $E = 200 \text{ kN/m}^2$. (06 Marks)

OR

- 10 a. Derive an expression for Euler's crippling load for both ends hinged column with usual notations. (08 Marks)
- b. Explain the failure of short column and long column. (04 Marks)
- c. Derive an expression for the slope and deflection for a simply supported beam carrying UDL over entire span. (08 Marks)

Third Semester B.E. Degree Examination, July/August 2022 Fluid Mechanics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Capillarity? Derive an expression for capillary rise of water in a glass tube. (06 Marks)
- b. A cube of 0.3m sides and weight 50N slides down an inclined plane sloped at 30° to the horizontal. The plane is covered by an oil of $\mu = 2.3$ Pa-s with 0.03mm thickness. Compute the velocity with which the cube slides down. (06 Marks)
- c. An inverted U – tube differential manometer is connected to two points of pipes A & B through which water is flowing. The vertical distance between the centres of these pipes is 30cm with B below A. Oil ($S = 0.8$) is used in manometer. The level of manometer liquid in the two limbs is 35cm above the centres of respective pipes. Determine the difference of pressure between the pipes. (08 Marks)

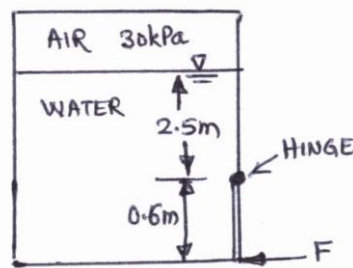
OR

- 2 a. Distinguish between : i) Ideal fluids and Real fluids
ii) Manometers and Pressure gauges iii) Surface tension and Capillarity. (06 Marks)
- b. State and prove Pascal's law. (06 Marks)
- c. An oil of viscosity 5 Poise is used for lubrication between a shaft and sleeve. The diameter of shaft is 50cm and it rotates at 200 rpm. Thickness of lubrication is 1mm. Calculate the power lost in overcoming the friction for a sleeve length of 100mm. (08 Marks)

Module-2

- 3 a. Define Velocity potential function and Stream function. Hence obtain Cauchy – Riemann equation. (08 Marks)
- b. Calculate the force F required to hold the hinged door shown in Fig. Q3(b) in closed position. The door is 0.6m square. Air pressure above water surface is 30 kPa. (12 Marks)

Fig. Q3(b)



OR

- 4 a. Derive an expression for total pressure and position of center of pressure on a vertically immersed plane surface. Show that center of pressure lies below CG. (10 Marks)
- b. For the two – dimensional flow defined by $u = x - 4y$ and $v = -y - 4x$. Obtain stream function and velocity potential function. (10 Marks)

Module-3

- 5 a. Define Impulse Momentum equation and give its applications. (04 Marks)
 b. Derive an expression for discharge through a horizontal venturimeter. (08 Marks)
 c. A pipeline carrying oil of specific gravity 0.8 changes in diameter from 300mm at A to 500mm at B which is 5m above A. The rate of flow is 200 ℓps and pressure at A and B is respectively 20N/cm^2 and 15N/cm^2 . Determine the head loss and direction of flow. (08 Marks)

OR

- 6 a. Derive an Euler's equation of motion along a stream line and obtain Bernoulli's equation. List assumptions. (10 Marks)
 b. A horizontal pipe of 300mm diameter is bent by 135° . If 250 ℓps of water is flowing through the pipe with a pressure of 400kPa, compute the magnitude and direction of force exerted on the bend. (10 Marks)

Module-4

- 7 a. Derive an expression for discharge through a small orifice. (08 Marks)
 b. Differentiate between : i) Orifice and mouthpiece ii) Notch and Weir. (04 Marks)
 c. Water is flowing in a rectangular channel 1m wide and 0.75m deep water. Find the discharge over a rectangular weir of 0.6m crest length with 200mm head over crest. Take $C_d = 0.65$. Consider velocity of approach and neglect end contraction. (08 Marks)

OR

- 8 a. Explain types of nappe, with neat sketches. (06 Marks)
 b. What is Cipoletti notch? What are its advantages? Explain with expression. (08 Marks)
 c. Compute the hydraulic coefficients of an orifice of 25mm diameter discharging under a constant head of 1.5m. The coordinates of jet from vena – contracta are (2.288m , 0.915m). The discharge measured is 102 ℓpm. (06 Marks)

Module-5

- 9 a. Derive Darcy – Weisbach equation for friction loss through a pipe. (08 Marks)
 b. Explain pipes in series and pipes in parallel. (06 Marks)
 c. A hydraulic pipeline 3km long and 500mm diameter is used to convey water at 1.5m/s velocity. Determine the magnitude of instantaneous pressure induced if the outflow valve is closed in i) 20 sec ii) 3.5 sec. Consider the pipe as rigid and take $K_{\text{water}} = 2\text{GPa}$. (06 Marks)

OR

- 10 a. Derive an expression for an instantaneous pressure induced in a pipe due to sudden closure of valve when pipe is rigid. (06 Marks)
 b. Explain Hardy Cross method. (06 Marks)
 c. A compound pipe in series consists of 1800m long of 0.5m diameter, 1200m long of 0.4m diameter and 600m long of 0.3m diameter connected between two tanks with difference in water levels of 100m. Determine the flow rate in the pipe neglecting minor losses. Also compute the diameter of equivalent pipe to be connected between the two tanks.

$$\text{Take } f = 0.04 \text{ in } h_f = \frac{fLV^2}{ZgD}.$$

(08 Marks)

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18CV34

Third Semester B.E. Degree Examination, July/August 2022 Building Materials and Construction

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Mention the importance of Stones, Bricks and Timber as construction materials. (05 Marks)
- b. Write a brief note on stabilized mud block. (05 Marks)
- c. List the different tests conducted on bricks. Explain any two of them. (10 Marks)

OR

- 2 a. Lists the tests conducted on fine aggregates and explain any one of them in detail. (08 Marks)
- b. What are the requirements of good building stones? (06 Marks)
- c. Explain the requirements of good bricks. (06 Marks)

Module-2

- 3 a. Briefly explain various methods of sub soil exploration. What are the factors on which the choice of a particular method depends? (10 Marks)
- b. Define Foundation. Explain different types of foundation with suitable sketches. (10 Marks)

OR

- 4 a. Write a note on classification of stone masonry. (06 Marks)
- b. Write a note on partitions walls. (04 Marks)
- c. Find dimension of a combined rectangular footing for two column A and B carrying a load of 500kN and 750kN column A 300×300 mm, column B 400×400 mm in size centre to centre of column is 3.4 mts SBC soil 150kN/m^2 . Width of footing is limited to 1.8m. (10 Marks)

Module-3

- Explain : i) Chejja ii) Canopy iii) Balcony iv) Lintel. (08 Marks)
- 5 a. Explain various modes failures of an arch. (08 Marks)
 - b. Highlight the important qualities of good flooring materials. (04 Marks)
 - c.

OR

- 6 a. What are the factors to be considered while selecting a roof covering? (05 Marks)
- b. Explain different types of lintels. (05 Marks)
- c. What are the requirements of good roof? Draw a neat sketch of king post truss and show important elements. (10 Marks)

Module-4

- 7 a. Explain briefly the guidelines to be followed while locating doors and windows. (10 Marks)
b. What are the requirements of good stair? (05 Marks)
c. Discuss importance of formwork in constructional activity. (05 Marks)

OR

- 8 a. Plan a dog legged stair for a building in which the vertical distance between the floors is 3.6m. The stair hall measure $2.5\text{m} \times 5\text{m}$. (10 Marks)
b. Write short notes on the following :
i) Ranking shore
ii) Underpinning by pit method. (10 Marks)

Module-5

- 9 a. What are the objectives of plastering? Discuss the defects in plastering. (10 Marks)
b. Explain the method of applying :
i) Stucco plastering
ii) Lathe plastering. (10 Marks)

OR

- 10 a. What are the defects in painting? (08 Marks)
b. Explain different types of plaster finishes. (06 Marks)
c. What are the characteristics of ideal paint? (06 Marks)

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Third Semester B.E. Degree Examination, July/August 2022

Basic Surveying

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Enumerate the applications of surveying in civil engineering. (06 Marks)
- b. Discuss the classification of surveying. (08 Marks)
- c. The distance between two points A and B measured along slope is 504 m. Find the horizontal distance between A and B when,
 - (i) The angle of slope is 12°
 - (ii) The slope is 1 in 4.5
 - (iii) The difference in Elevation of A and B is 65 m (06 Marks)

OR

- 2 a. What is field book? List the points to be kept in mind while entering in field book. (08 Marks)
- b. A rectangular plot measures $20\text{cm} \times 30\text{cm}$ on a village map drawn to a scale of $1\text{ cm} = 100\text{ m}$. Calculate area in hectares. If it is redrawn on topo sheet to scale of $1\text{ km} = 1\text{ cm}$. What will be its area? Determine representative fraction in both the cases. (06 Marks)
- c. A survey line BAC crosses a river, A and C being on the near and distant banks respectively. A 50 m perpendicular line to AB is measured from "A". Bearing of "C" and "B" are 320° and 230° respectively. AB being 25 m. Find the width of the river. (06 Marks)

Module-2

- 3 a. Differentiate between :
 - (i) True meridian and magnetic meridian. (ii) WCB and QB
 - (iii) Fore bearing and back bearing. (06 Marks)
- b. Compute the bearings for setting out regular pentagon, if the bearing of line AB is $30^\circ 0'$. (06 Marks)
- c. The following bearing were observed in running a closed traverse. Determine the correct magnetic bearings of the lines. (08 Marks)

Line	FB	BB
AB	$38^\circ 30'$	$219^\circ 15'$
BC	$100^\circ 45'$	$278^\circ 30'$
CD	$25^\circ 45'$	$207^\circ 30'$
DE	$325^\circ 15'$	$145^\circ 15'$
EA	$190^\circ 30'$	$10^\circ 15'$

OR

- 4 a. Explain the following :
 - (i) Latitude and Departure (ii) Dependent and Independent co-ordinates (06 Marks)
 - b. What is closing error? Explain Bowditch's method of adjusting the traverse. (06 Marks)
 - c. Adjust the following traverse using transit rule: (08 Marks)

Line	AB	BC	CD	DE	EA
Latitude	62.96	67.60	-143.67	-104.97	118.57
Departure	63.33	209.10	47.05	-119.55	-199.70

Module-3

- 5 a. Write a short note on Auto level. (05 Marks)
 b. Define the following terms:
 (i) Reduced level. (ii) Turning point (iii) MSL
 (iv) Back sight (v) H.I. (05 Marks)
 c. The following staff readings were observed successively with a level, the instrument having been moved after third, sixth and eighth readings. Enter readings and calculate RL of points by H.I method, if first reading was taken with a staff held on BM = 432.384 m.
 2.228 m, 1.606, 0.988, 2.090, 2.864, 1.262, 0.602, 1.982, 1.044, 2.684 m (10 Marks)

OR

- 6 a. Explain differential leveling with a neat sketch. (06 Marks)
 b. Write a note on curvature and refraction correction. (06 Marks)
 c. Following details were recorded in level work. Calculate (i) True R.L of point "B"
 (ii) Angular error in collimation (iii) Combined correction for curvature and refraction.
 If distance between A and B is 1000 m. (08 Marks)

Instrument at	Staff reading on		Remarks
	A	B	
A	1.625	2.545	R.L.A = 100.80 m
B	0.725	1.405	

Module-4

- 7 a. What are the advantages and disadvantages of plane table surveying? (06 Marks)
 b. Describe briefly radiation method and intersection method of plane tabling. (10 Marks)
 c. Write short notes on orientation of plane table. (04 Marks)

OR

- 8 a. Explain the graphical method of solution of two point problem with sketches. (08 Marks)
 b. Describe the different forms of errors in plane table. (06 Marks)
 c. Briefly explain the working operation of plane table surveying. (06 Marks)

Module-5

- 9 a. Define contour. List any six characteristics of a contour with sketches. (08 Marks)
 b. Discuss the different methods of determining area. (04 Marks)
 c. A series of offsets were taken from a chain line to a curved boundary line at 10 m intervals in the following order: 3.25, 5.60, 4.20, 6.65, 8.75, 6.20, 3.25, 4.20, 5.65
 Calculate the area by,
 (i) Average ordinate rule. (ii) Trapezoidal rule
 (iii) Simpson's rule (08 Marks)

OR

- 10 a. A road embankment is 30 m wide at the top with side slopes of 2 : 1. The ground levels at 100 m intervals along a line AB are as under:
 A170.30, 169.10, 168.50, 168.10, 166.50B.
 The formation level at "A" is 178.70 m, with uniform falling gradient of 1 in 50 from A to B. Determine the volume of earth work by prismoidal rule. Assume the ground to be in cross section. (08 Marks)
 b. Explain Interpolation of contours. List the methods of contouring. (06 Marks)
 c. Write short notes on Digital planimeter. (06 Marks)

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Third Semester B.E. Degree Examination, July/August 2022 Engineering Geology

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain briefly the Internal structure of the Earth. (08 Marks)
- b. Explain the role of Geology in the field of Civil Engineering. (08 Marks)
- c. Name the different branches of Geology which are related to Civil Engineering. (04 Marks)

OR

- 2 a. Define a Mineral. Briefly explain the physical properties of a Mineral. (12 Marks)
- b. Describe any two of the following Minerals : (08 Marks)
 - i) Quartz ii) Hematite iii) Pyrite iv) Mica.

Module-2

- 3 a. Explain the different forms of Igneous Rocks with neat sketches. (12 Marks)
- b. Enumerate the classification of Igneous Rocks. (08 Marks)

OR

- 4 a. What is Sedimentation? Explain the Primary Sedimentary structures, with neat sketches. (10 Marks)
- b. What is Metamorphism? Explain the different kinds of Metamorphism with example. (10 Marks)

Module-3

- 5 a. Define the terms Dip , Strike and Outcrop. (06 Marks)
- b. Explain with a neat sketch, Mural Joints and Columnar Joints. (06 Marks)
- c. What are Folds? Explain briefly the different types of folds with neat sketches. (08 Marks)

OR

- 6 a. What are Faults? Name different types of faults. What are the Engineering considerations of faults in the Civil Engineering Projects? (10 Marks)
- b. What are the factors considered for the selection of site for dams and reservoirs? (10 Marks)

Module-4

- 7 a. Explain the vertical distribution of ground water with a neat sketch. (10 Marks)
- b. Define Aquifer. Explain briefly the different types of Aquifer, with a neat sketch. (10 Marks)

OR

- 8 a. Explain the ground water exploration by Electrical Resistivity method. (10 Marks)
- b. Define the terms : (06 Marks)
 - i) Porosity ii) Permeability iii) Transmissibility.
- c. Write short note on Flood control. (04 Marks)

Module-5

- 9 a. What are Earthquake? Add a note on causes and effects of Earthquake. (10 Marks)
b. Write short notes on :
i) Tsunami – causes and effects.
ii) Landslides – Causes and remedial measures. (10 Marks)

OR

- 10 a. What is Remote Sensing? What are its application in Civil Engineering? (08 Marks)
b. What is GIS? Name the different components of GIS. (06 Marks)
c. What are the applications of GPS in Civil Engineering? (06 Marks)

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Third Semester B.E. Degree Examination, July/August 2022

Additional Mathematics – I

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Express $\frac{(3+i)(1-3i)}{(2+i)}$ in the form $x + iy$. (06 Marks)
- b. If $\vec{a} = i - 2j + 3k$, $\vec{b} = -i + 2j + k$ and $\vec{c} = 3i + j$. Find the value of 'p' such that $\vec{a} - p\vec{b}$ is perpendicular to $\vec{c}$. (07 Marks)
- c. Find the angle between the vector $\vec{a} = 5\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} + 6\hat{k}$. (07 Marks)

OR

- 2 a. Find the modulus and amplitude of the complex number $1 + \cos\alpha + i \sin\alpha$. (06 Marks)
- b. Prove that $\left(\frac{1 + \cos\theta + i \sin\theta}{1 + \cos\theta - i \sin\theta} \right)^n = \cos n\theta + i \sin n\theta$. (07 Marks)
- c. Find the sine of the angle between $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$. (07 Marks)

Module-2

- 3 a. Find the n^{th} derivative of $\cos x \cos 2x$. (06 Marks)
- b. Obtain the Maclaurin's series expansion of the function $\sqrt{1 + \sin 2x}$ upto the term containing x^4 . (07 Marks)
- c. If $u = f(y - z, z - x, x - y)$ prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. (07 Marks)

OR

- 4 a. If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$ prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$. (06 Marks)
- b. If $z = xy^2 + x^2y$ where $x = at^2$ and $y = 2at$. Find $\frac{dz}{dt}$. (07 Marks)
- c. If $x = e^u \sec v$, $y = e^u \tan v$. Find $J \begin{pmatrix} x, y \\ u, v \end{pmatrix}$. (07 Marks)

Module-3

- 5 a. A particle moves along the curve $\vec{r} = \cos 2t\hat{i} + \sin 2t\hat{j} + t\hat{k}$ where t is the time variable. Determine the components of velocity and acceleration vectors at $t = \pi/8$ in the direction of $\sqrt{2}\hat{i} + \sqrt{2}\hat{j} + \hat{k}$. (06 Marks)
- b. Find $\text{div } f$ for $f = \nabla(x^3 + y^3 + z^3 - 3xyz)$. (07 Marks)
- c. Show that $f = (2xy + z^2)\hat{i} + (x^2 + 2yz)\hat{j} + (y^2 + 2xz)\hat{k}$ is irrotational and find ϕ such that $f = \nabla\phi$. (07 Marks)

OR

- 6 a. Find the unit normal to the surface $x^3y^3z^2 = 4$ at the point $P(-1, -1, 2)$. (06 Marks)
- b. If $f = 2x^2\hat{i} - 3yz\hat{j} + xz^2\hat{k}$ and $\phi = 2z - x^3y$, find $f \cdot (\nabla\phi)$ and $f \times (\nabla\phi)$ at $(1, -1, 1)$. (07 Marks)
- c. Show that $\vec{f} = \frac{x\hat{i} + y\hat{j}}{x^2 + y^2}$ is both solenoidal and irrotational. (07 Marks)

Module-4

- 7 a. Obtain a reduction formula for $\int_0^{\pi/2} \sin^n x \, dx$ ($n > 0$). (06 Marks)
- b. Evaluate $\int_0^{2a} x^2 \sqrt{2ax - x^2} \, dx$. (07 Marks)
- c. Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz \, dz \, dy \, dx$. (07 Marks)

OR

- 8 a. Obtain a reduction formula for $\int_0^{\pi/2} \cos^n x \, dx$ ($n > 0$). (06 Marks)
- b. Evaluate $\iint_R xy \, dx \, dy$ where R is the first quadrant of the circle $x^2 + y^2 = a^2$, $x \geq 0$, $y \geq 0$. (07 Marks)
- c. Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x + y + z) \, dy \, dx \, dz$. (07 Marks)

Module-5

- 9 a. Solve $x^2 \frac{dy}{dx} - 2xy - x + 1 = 0$. (06 Marks)
- b. Solve $(3x^2y^2 + x^2)dx + (2x^3y + y^2)dy = 0$. (07 Marks)
- c. Solve $3x(x + y^2)dy + (x^3 - 3xy - 2y^3)dx = 0$. (07 Marks)

OR

- 10 a. Solve $\left[y \left(1 + \frac{1}{x} \right) + \cos y \right] dx + [x + \log x - x \sin y] dy = 0$. (06 Marks)
- b. Solve $\frac{dy}{dx} + y \cot x = \sin x$. (07 Marks)
- c. Solve $\frac{dy}{dx} + \frac{y}{x} = y^2x$. (07 Marks)

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CBCS SCHEME

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18MAT41

Fourth Semester B.E. Degree Examination, July/August 2022 Complex Analysis, Probability and Statistical Methods

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Derive Cauchy-Riemann equation in Polar form. (06 Marks)
- b. Find the analytic function $f(z)$ whose real part is $x \sin x \cosh y - y \cos x \sinh y$ (07 Marks)
- c. If $f(z)$ is analytic show that $\left| \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right| |f(z)|^2 = 4 |f'(z)|^2$ (07 Marks)

OR

- 2 a. Find the analytic function $f(z)$ given that the sum of its real and imaginary part is $x^3 - y^3 + 3xy(x - y)$ (06 Marks)
- b. Find the analytic function $f(z) = u + iv$ if $v = r^2 \cos 2\theta - r \cos \theta + 2$ (07 Marks)
- c. If $f(z)$ is analytic function then show that $\left\{ \frac{\partial}{\partial x} |f(z)| \right\}^2 + \left\{ \frac{\partial}{\partial y} |f(z)| \right\}^2 = |f'(z)|^2$ (07 Marks)

Module-2

- 3 a. State and prove Cauchy's Integral formula. (06 Marks)
- b. Evaluate $\int_0^{2+i} \frac{1}{z^2} dz$ along (i) the line $y = \frac{x}{2}$ (ii) The real axis to 2 and then vertically to $2 + i$. (07 Marks)
- c. Find the bilinear transformation which maps the points 1, i , -1 onto the points i , 0, $-i$ respectively. (07 Marks)

OR

- 4 a. Discuss the transformation $w = e^z$, with respect to straight lines parallel to x and y axis. (06 Marks)
- b. Using Cauchy's integral formula evaluate $\int_c \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where $c: |z| = 3$ (07 Marks)
- c. Find the bilinear transformation which maps the points 0, 1, ∞ on to the points -5 , -1 , 3 respectively. (07 Marks)

Module-3

- 5 a. A random variable X has the following probability function for various values of X .

X	0	1	2	3	4	5	6	7
P(X)	0	k	2k	2k	3k	k ²	2k ²	7k ² +k

Find i) k ii) $P(X < 6)$ iii) $P(3 < X \leq 6)$

(06 Marks)

- b. Out of 800 families with 5 children each, how many families would you expect to have
(i) 3 boys (ii) 5 girls (iii) either 2 or 3 boys (iv) atmost 2 girls, assuming equal probabilities for boys and girls. (07 Marks)

- c. The length in time (minutes) that a certain lady speaks on a telephone is a random variable with probability density function

$$f(x) = \begin{cases} Ae^{-x/5} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Find the value of the constant A. What is the probability that she will speak over the phone for (i) More than 10 minutes (ii) Less than 5 minutes (iii) Between 5 and 10 minutes.

(07 Marks)

OR

- 6 a. Find the constant C such that the function

$$f(x) = \begin{cases} Cx^2, & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases} \quad \text{is a probability density function. Also compute } P(1 < x < 2),$$

$P(x \leq 1)$ and $P(x > 1)$ (06 Marks)

- b. 2% fuses manufactured by a firm are found to be defective. Find the probability that the box containing 200 fuses contains

(i) No defective fuses (ii) 3 or more defective fuses (iii) At least one defective fuse.

(07 Marks)

- c. If x is a normal variate with mean 30 and standard deviation 5 find the probabilities that
(i) $26 \leq x \leq 40$ (ii) $x \geq 45$ (iii) $|x - 30| > 5$

Given that $\phi(1) = 0.3413$, $\phi(0.8) = 0.2881$, $\phi(2) = 0.4772$, $\phi(3) = 0.4987$ (07 Marks)

Module-4

- 7 a. The following table gives the ages (in years) of 10 married couples. Calculate Karl Pearson's coefficient of correlation between their ages:

Age of husband (x)	23	27	28	29	30	31	33	35	36	39
Age of wife (y)	18	22	23	24	25	26	28	29	30	32

(06 Marks)

- b. In a partially destroyed laboratory record of correlation data only the following results are available:

Variance of x is 9 and regression lines are $8x - 10y + 66 = 0$, $40x - 18y = 214$. Find

- (i) Mean value of x and y
(ii) Standard deviation of y
(iii) Coefficient of correlation between x and y.

(07 Marks)

- c. Fit a parabola of the form $y = ax^2 + bx + c$ for the data

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

(07 Marks)

OR

- 8 a. Obtain the lines of regression and hence find the coefficient of correlation of the data:

x	1	3	4	2	5	8	9	10	13	15
y	8	6	10	8	12	16	16	10	32	32

(06 Marks)

- b. Show that if θ is the angle between the lines of regression

$$\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1 - r^2}{r} \right)$$

(07 Marks)

- c. Fit a straight line $y = a + bx$ to the data

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

(07 Marks)

Module-5

- 9 a. The joint probability distribution of the random variables X and Y is given below.

X \ Y	-4	2	7
1	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$
5	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$

Find (i) $E[X]$ and $E[Y]$ (ii) $E[XY]$ (iii) $\text{cov}(X, Y)$ (iv) $\rho(X, Y)$.

Also, show that X and Y are not independent.

(06 Marks)

- b. A manufacturer claimed that atleast 95% of the equipment which he supplied to a factory confirmed to specifications. An examination of a sample of 200 pieces of equipment revealed that 18 of them were faulty. Test his claim at a significance level of 1% and 5% ($z_{0.05} = 1.96$, $z_{0.01} = 2.58$). (07 Marks)
- c. A certain stimulus administered to each of the 12 patients resulted in the following change in blood pressure 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4. Can it be concluded that the stimulus will increase the blood pressure ($t_{0.05}$ for 11 d.f. is 2.201) (07 Marks)

OR

- 10 a. Define the terms :

(i) Null hypothesis (ii) Type-I and Type – II errors (iii) Significance level (06 Marks)

- b. In an experiment of pea breeding the following frequencies of seeds were obtained:

Round Yellow	Wrinkled Yellow	Round Green	Wrinkled Green	Total
315	101	108	32	556

Theory predicts that the frequencies should be in proportions 9:3:3:1

Is the experiment in agreement with theory ($\chi^2_{0.5}$ for 3 d.f is 7.815)

(07 Marks)

- c. The joint probability distribution of two discrete random variable X and Y is given by $f(x, y) = k(2x + y)$ where x and y are integers such that $0 \leq x \leq 2$, $0 \leq y \leq 3$. Find k and the marginal probability distribution of X and Y. Show that the random variables X and Y are dependent. Also, find $P(X \geq 1, Y \leq 2)$. (07 Marks)

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18CV42

Fourth Semester B.E. Degree Examination, July/August 2022 Analysis of Determinate Structures

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain with examples statically determinate and indeterminate structures. (08 Marks)
b. Find the Static and Kinematic indeterminacies of the following structures.

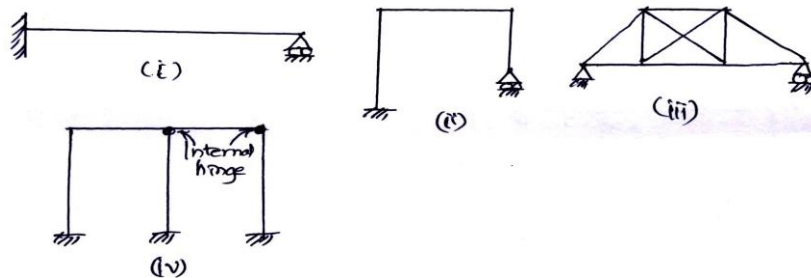


Fig.Q1(b)

(12 Marks)

OR

- 2 a. What do you mean by influence line diagram and state its applications. (08 Marks)
b. Draw ILD for
(i) Reactions at supports of a simply supported beam.
(ii) Shear force of a simply supported beam carrying concentrated unit load. (12 Marks)

Module-2

- 3 a. Two point loads 4 kN and 6 kN spaced 6m apart cross a girder of 16m span, the 4 kN load, leading from left to right. Construct the maximum SF and BM diagrams stating the absolute maximum values. [Fig.Q3(a)].



Fig.Q3(a)

(10 Marks)

- b. Draw the influence line for SF and BM at a section 5m from the left hand support of a simply supported beam 25m span. Hence calculate maximum shear force and BM at this section due to uniformly distributed load of 1 kN/m, 8m long. [Refer Fig.Q3(b)]

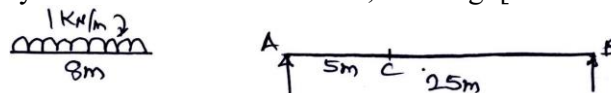


Fig.Q3(b)

(10 Marks)

OR

- 4 A simply supported beam of span 20m is subjected to a set of loads of magnitude of 20 kN, 30 kN, 15 kN and 10 kN spaced as shown with 10 kN leading. Determine the maximum BM at a section 5m from the left end and also the absolute maximum BM developed in the beam. [Refer Fig.Q4] (20 Marks)

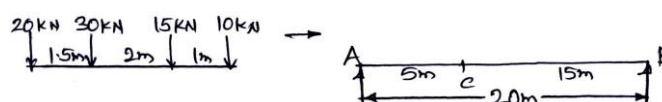


Fig.Q4

Module-3

- 5 a. Determine the slope and deflection at the free end of a cantilever beam loaded as shown in the Fig.Q5(a). Take $EI = 4 \times 10^5 \text{ kNm}^2$. Use moment area method.

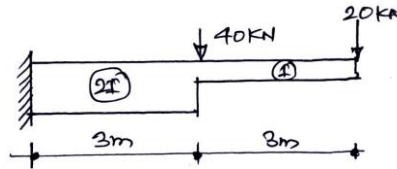


Fig.Q5(a)

(10 Marks)

- b. Determine the slope at C and deflection at D of a simply supported beam shown in Fig.Q5(b). Take $E = 200 \text{ GPa}$, $I = 2 \times 10^6 \text{ mm}^4$. Use conjugate beam method.

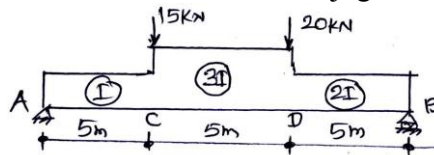


Fig.Q5(b)

(10 Marks)

OR

- 6 a. Determine the slope at the supports and deflection at the centre of a simply supported beam with a point load W at its mid span. Use moment area method. (10 Marks)
- b. Determine the slope at the supports and deflection at the centre of a simply supported beam with uniformly distributed load of W/m over the entire span. Use moment area method. (10 Marks)

Module-4

- 7 a. Derive the expression for strain energy stored in an prismatic element subjected to pure bending moment. (08 Marks)
- b. Determine the vertical deflection at C of a bent frame shown in the Fig.Q7(b). Use Castigliano's approach. Take $E = 200 \text{ GPa}$, $I = 80 \times 10^7 \text{ mm}^4$.

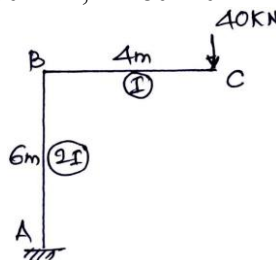


Fig.Q7(b)

(12 Marks)

OR

- 8 Determine the vertical and horizontal deflection of the point C, of the pin jointed frame shown in Fig.Q8. The cross sectional area of AB = 100 sqmm and BC and AC are 150 sqmm. Take $E = 2 \times 10^5 \text{ N/mm}^2$.

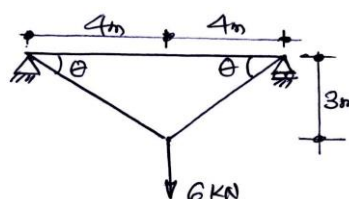


Fig.Q8

(20 Marks)

Module-5

- 9 A three hinged parabolic arch of 20m span with 4m central rise carries a point load of 4kN at 4m horizontally from the left hinge. Calculate the normal thrust and radial shear at a section just after the load. Also calculate the maximum positive and negative BM. Sketch BMD.

(20 Marks)**OR**

- 10 A cable is of uniform section is suspended between two supports 100m apart. It carries a uniformly distributed load of 10 kN/m spread over the horizontal span. Find

- (i) Maximum and minimum tension in the cable.
- (ii) Minimum cross sectional area of the cable required if the allowable stress is 300 MPa.
- (iii) Length of the cable.

(20 Marks)

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GBCS SCHEME

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18CV43

Fourth Semester B.E. Degree Examination, July/August 2022 Applied Hydraulics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define repeating variables. What are the guidelines for selecting repeating variables? (06 Marks)
- b. Derive the scale ratios for the following quantities as per Froude model law:
(i) Velocity (ii) Discharge (iii) Force (iv) Pressure (08 Marks)
- c. A spillway model is constructed in the laboratory such that the velocity and discharge in the model are respectively 2 m/s and 2.5 m³/s. If the velocity in the prototype is 40 m/s, what is the scale ratio and the discharge in the prototype? (06 Marks)

OR

- 2 a. Define: (i) Meta center (ii) Metacentric height. Explain how these are used to check the stability of floating bodies. (06 Marks)
- b. The resisting torque T against the motion of a shaft in a lubricated bearing depends on viscosity μ , the rotational speed N , the diameter D and bearing pressure p . Show that $T = \mu ND^3 \cdot \phi \left[\frac{p}{\mu N} \right]$ where ϕ represents function. (08 Marks)
- c. Find the volume of water displaced and position of center of buoyancy for a wooden block of density 650 kg/m³ when it floats in water horizontally. The width of block is 2.5 m, depth is 1.5 m and length is 6m. (06 Marks)

Module-2

- 3 a. Distinguish between:
(i) Gradually varied flow and rapidly varied flow
(ii) Total energy and specific energy
(iii) Subcritical flow and supercritical flow (06 Marks)
- b. The discharge of water through a rectangular channel of 8m width is 16 m³/s when the depth of water is 1.2 m. Calculate:
(i) Specific energy
(ii) Critical depth and critical velocity
(iii) Minimum specific energy (06 Marks)
- c. For most economical trapezoidal section, prove that half of top width is equal to side slope length. Also show that hydraulic mean depth is equal to half of the flow depth. (08 Marks)

OR

- 4 a. List the conditions at critical point on specific energy curve. (04 Marks)
- b. Derive Chezy's equation for uniform rate of flow in a channel. Hence write Manning's equation. (08 Marks)
- c. A rectangular channel 6m wide and 1m depth of water has a bed slope of 1 in 900 and $n = 0.012$. Determine the discharge. What will be the dimensions of this channel for maximum discharge? Also compute percentage increase in discharge. (08 Marks)

Module-3

- 5 a. Define hydraulic jump. List the application of hydraulic jump. (04 Marks)
 b. Derive dynamic or differential equation for GVF. (08 Marks)
 c. A rectangular channel 8m wide discharges water with a depth of 0.4 m. If the critical depth of flow is 0.8 m, compute the depth of flow after jump, if jump occurs. Further, also calculate energy loss. (08 Marks)

OR

- 6 a. Give the flow depth condition for M1, S3, C1, H2 water surface profiles. (04 Marks)
 b. Derive the relationship between conjugate depths in case of a hydraulic jump in a rectangular channel. (08 Marks)
 c. A rectangular channel with bottom width 4m and bed slope 0.0008 has a discharge of 1.5 m³/s. The depth of flow at a certain section in GVF is 0.5 m. If $n = 0.016$, identify type of profile. (08 Marks)

Module-4

- 7 a. Define efficiencies of turbines and give relationship between them. (05 Marks)
 b. Show that the maximum efficiency of jet striking at the center of series of symmetrical curves vanes is $\eta_{\max} = \frac{1}{2}(1 + \cos\theta)$ (08 Marks)
 c. A pelton wheel turbine has to be designed for the following specification:
 Shaft power = 12000 KW, Head = 380 m, speed = 750 rpm, $\eta_0 = 85\%$, $d = \frac{D}{10}$, $C_v = 0.98$, $C_u = 0.45$. Determine: (i) Runner diameter (ii) Jet diameter (iii) Number of jets (07 Marks)

OR

- 8 a. State impulse-momentum equation. Give its applications. (04 Marks)
 b. Sketch and explain general layout of hydroelectric power plant. (06 Marks)
 c. A jet of water moving at 15 m/s impinges on a symmetrically curved moving vane tangentially to deflect the jet through 120°, the velocity of vane is 5 m/s. Find the angle of jet so that there is no shock. What is the absolute velocity of jet at exit in magnitude and direction? Also compute work done per N of water. (10 Marks)

Module-5

- 9 a. Define draft tube. List its functions. (04 Marks)
 b. Compute the minimum starting speed of a centrifugal pump having 90 cm and 45 cm as impeller diameters at outlet and inlet working against 10m head. (06 Marks)
 c. The inner and outer diameter of an inward flow Francis turbine are 0.6 m and 1 m, respectively. The width at both inlet and outlet is 12 cm. The head over the turbine is 9m. Hydraulic efficiency = 90%, Vane angle at outlet is 20° and discharge at outlet is radial with flow velocity of 2.7 m/s. Compute:
 (i) Guide blade angle (ii) Vane angle at inlet (iii) Speed of turbine
 (iv) Discharge (v) Water power (10 Marks)

OR

- 10 a. Explain with sketch various components of a centrifugal pump. (07 Marks)
 b. A centrifugal pump is to delivery 0.15 m³/s at a speed of 1450 rpm against a head of 25 m. The impeller diameter is 250 mm and width is 50 mm at outlet. $\eta_{\text{man}} = 75\%$. Determine the vane angle at the outer periphery of impeller. (08 Marks)
 c. Write a note of multistage centrifugal pumps. (05 Marks)

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Fourth Semester B.E. Degree Examination, July/August 2022

Concrete Technology

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Mention different types of cement. (10 Marks)
- b. Explain the following : (10 Marks)
 - i) Fly ash ii) Silico Fumes iii) Rice husk ash iv) GGBS.

OR

- 2 a. Explain the constituents of cement with their percentages functioning. (10 Marks)
- b. Explain the importance of size shape and texture of coarse aggregate on quality of concrete. (10 Marks)

Module-2

- 3 a. List and explain the factors affects the workability of concrete. (10 Marks)
- b. Enumerate good and bad practices of making and using of Fresh concrete. (10 Marks)

OR

- 4 a. Explain the process of hydration of cement. Enumerate its significance and the chemical reaction. (10 Marks)
- b. Write short notes : (10 Marks)
 - i) Segregation ii) Membrane curing.

Module-3

- 5 a. Explain the factors affecting the strength of concrete. (10 Marks)
- b. What is the Necessity of Non Destructive Testing (NDT)? Explain any two methods of NDT. (10 Marks)

OR

- 6 a. Mention the types of Shrinkage. Explain factor affecting Shrinkage. (10 Marks)
- b. Explain Sulphate attack and chloride attack. (10 Marks)

Module-4

- 7 a. What are the objectives of mix design? Explain the factors to be considered for mix design. (10 Marks)
- b. Mention different method of mix design; explain the factor affecting the choice of mix proportions. (10 Marks)

OR

- 8 Design a concrete mix of M₂₅ grade as per IS 10262-2019, with the following stipulations.
- Grade designation – M₂₅
 - Type of cement OPC 43 grade
 - Maximum Nominal size of aggregate 20mm down
 - Minimum cement content : 300 Kg/m³
 - Workability : Slump : 75mm
 - Exposure condition : moderate
 - Method of concrete placing : Manual
 - Maximum cement content : 450Kg/m³
 - Chemical admixture : NIL
 - Fine aggregate zone : zone 2
- A. Specific gravity of cement : 3.15
- B. Coarse aggregate
- Specific gravity 2.80
 - Water absorption 1%
 - Free surface moisture content : NIL
- C. Fine aggregate
- Specific gravity 2.65
 - Water absorption 2% Free surface moisture content : 2% (20 Marks)

Module-5

- 9 a. Enumerate the necessity of RMC with advantages and disadvantages. (10 Marks)
- b. Write short notes on :
- Geopolymer concrete
 - High performance concrete. (10 Marks)

OR

- 10 a. List the types of Fibres used in FRC and discuss factors affecting properties of FRC. (10 Marks)
- b. What is self compacting concrete? Explain the properties of SCC. (10 Marks)

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18CV45

Fourth Semester B.E. Degree Examination, July/August 2022

Advanced Surveying

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the procedure of measuring horizontal angle by Repetition and Reiteration method. (08 Marks)
- b. Define the following terms:

(i) Centring

(ii) Transiting

(iii) Changing face

(iii) Swinging

(v) Line of collimation

(06 Marks)
- c. Write the short notes on step by step procedure for obtaining permanent adjustment of transit theodolite. (06 Marks)

OR

- 2 a. To find the elevation of the top (Q) of a hill, a flag staff of 2 m height was erected and observations were made from 2 stations P and R, 60 m apart. The horizontal angle measured at P between R and the top of the flat staff was $60^{\circ} 30'$ and that measured at R between the top of the flags staff and P was $68^{\circ} 18'$. The angle of elevation to the top of the flag staff was measured to be $10^{\circ} 12'$ at P. The angle of elevation to the top of the flag staff was measured to be $10^{\circ} 48'$ at R. Staff readings on B.M. when the instrument was at P = 1.965 m and that with the instrument at R = 2.055 m. Calculate the elevation of the top of the hill if that of B.M. was 435.065 m. (10 Marks)
- b. Derive expression for determining the distance and elevation of an inaccessible object by single plane method. Assume instrument axes at the same level and different level. (10 Marks)

Module-2

- 3 a. Derive distance and elevation formulae for stadia tachometry, when staff held normal to the line of sight, for both an angle of elevation and angle of depression. (10 Marks)
- b. Determine the gradient from point A to a point B from the following observations made with a tacheometer fitted with an anallactic lens. The constant of the instrument was 100 and the staff was held vertically,

Inst Station	Staff point	Bearing	Vertical angle	Staff readings
P	A	134°	$+10^{\circ} 32'$	1.360, 1.915, 2.470
	B	224°	$+5^{\circ} 36'$	1.065, 1.885, 2.705

(10 Marks)

OR

- 4 a. List the various factors that are to be considered in the selection of site for baseline in triangulation survey. (04 Marks)
- b. Write the objectives of Geodetic Triangulation and briefly explain the various triangulation figures adopted in triangulation survey with neat sketch. (10 Marks)
- c. From an eccentric station S, 12.25 m to the west of the main station B, the following angles were measured $\angle BSC = 76^{\circ} 25' 32''$, $\angle CSA = 54^{\circ} 32' 20''$. The stations S and C are to the opposite sides of the line AB. Calculate the correct angle ABC if the lengths AB and BC are 5286.5 m and 4932.2 m respectively. (06 Marks)

Module-3

- 5 a. With the help of neat sketch of a simple circular curve, explain (i) point of curve (ii) deflection angle (iii) tangent length (iv) length of the long chord (v) Apex distance (vi) Mid ordinate (06 Marks)
- b. Two tangents AB and BC intersect at a point B at chainage 2760 m. Calculate all the necessary data for setting out a curve of 573 m radius and deflection angle 30° by the method of ordinates from long chord at 30 m interval. (08 Marks)
- c. Two straights AB and BC are intersected by a line D_1D_2 . The angles AD_1D_2 and CD_2D_1 are $139^\circ 30'$ and $143^\circ 36'$ respectively. The radius of the first arc is 600 m and that of the second arc is 800 m. If the chainage of the second arc is 800 m. If the chainage of Intersection point B is 8248.1. Find the chainage of the tangent points and the Point of Compound Curvature (P.C.C). (06 Marks)

OR

- 6 a. Two parallel railway lines are to be connected by a reverse curve, each section having the same radius. If the lines are 12 m apart and the maximum distance between tangent points measured parallel to the straights is 48 m, find the maximum allowable radius. If however, both the radius are to be different, calculate the radius of the second branch if that of the first branch is 60 m. Also calculate the length of both the branches. (10 Marks)
- b. What are the requirements of an essential transition curve? (06 Marks)
- c. With a neat sketch, list any four vertical curves. (04 Marks)

Module-4

- 7 a. A vertical photograph was taken at an altitude of 1200 m above mean sea level. Determine the scale of the photograph for terrain lying at elevations of 80 m and 300 m if the focal length of the camera is 15 cm. (08 Marks)
- b. Derive the expression for relief displacement on a vertical photograph. (06 Marks)
- c. List the reasons for keeping overlap in photographs. (06 Marks)

OR

- 8 a. Define (i) Vertical photograph (ii) Tilted photograph (iii) Oblique photograph. (06 Marks)
- b. Write short note on : (i) Stereoscope (ii) Parallax Bar (iii) Mosaics (08 Marks)
- c. The scale of an aerial photograph is 1 cm = 160 m and the size of the photograph is 20 cm × 20 cm. If the longitudinal lap is 65% and side lap is 35%, determine the number of photographs required to cover an area of 232 sq.km. (06 Marks)

Module-5

- 9 a. Mention the advantages of total station and describe its working principle. (08 Marks)
- b. With neat sketch, explain the electromagnetic spectrum. (06 Marks)
- c. Explain the interaction of electromagnetic waves with atmosphere. (06 Marks)

OR

- 10 a. Define Remote Sensing? Explain the stages of idealized remote sensing system. (08 Marks)
- b. With a neat sketch, explain the components of GIS. (06 Marks)
- c. List out the application of GIS and Remote Sensing. (06 Marks)

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18CV46

Fourth Semester B.E. Degree Examination, July/August 2022

Water Supply and Treatment Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the various types of water demand. (06 Marks)
- b. Discuss the factors that affect per capita demand. (08 Marks)
- c. Calculate the fire demand for 2 lakh population by different formula. (06 Marks)

OR

- 2 a. What is population forecasting? List any eight methods of population forecasting. Explain the incremental increase and comparative graphical method. (10 Marks)
- b. The following is the population data of a city available from past census records. Determine the future population of the city in 2040 by
 - i) Arithmetical increase method
 - ii) Geometrical increase method.

Year	1970	1980	1990	2000	2010
Population	25,000	28,000	34,000	42,000	47,000

(10 Marks)

Module-2

- 3 a. Give complete sequence of a water treatment plant with a flow diagram and mention the significance of each unit. (08 Marks)
- b. Briefly explain different methods of water sampling. (06 Marks)
- c. Give the drinking water quality standards as per BIS for the following parameters. Discuss their effects when they are exceeding their limits. i) Iron ii) Chlorides iii) Fluoride iv) Nitrate. (06 Marks)

OR

- 4 a. Explain the factors to be considered while selecting a site for intake structures. (06 Marks)
- b. What are intake structures? List the different types of intakes and explain any one, with neat sketch. (08 Marks)
- c. List the different types of aeration process and explain any one, with neat sketch. (06 Marks)

Module-3

- 5 a. Briefly explain the term i) Sedimentation ii) Coagulation iii) Clarification. (06 Marks)
- b. How you will determine the optimum coagulant dosage in lab using Jar test apparatus? Discuss with sketch. (07 Marks)
- c. The maximum daily demand at a water purification tank plant is 12MLD. Design the dimensions of a suitable rectangular sedimentation tank for the raw water supplies. Take detention time period of 6 hours and the depth of 4.0mts. The velocity of flow is 20cm/min. (07 Marks)

OR

- 6 a. Briefly explain theory of filtration. (06 Marks)
 b. With the help of a neat sketch, explain the working of rapid gravity filter. (08 Marks)
 c. Determine the dimensions of a set of rapid gravity filters for treating water required for a population of 50,000 with average rate of demand as 180 litres per day per person. Assume a peak factor of 1.8 by ignoring wash water requirements. Assume rate of filtration is 5000l/h/sq.m. (06 Marks)

Module-4

- 7 a. Explain theory of chlorination of water with chemical equation. (06 Marks)
 b. Explain break point chlorination graphically. (06 Marks)
 c. Estimate the quantity a bleaching powder required for a city water tank, for one month (31 days), use the following data:
 Population: 1 lakh
 Strength of bleaching powder: 20%
 Chlorine demand of water: 0.4mg/l
 Residual chlorine required: 0.2mg/l
 Rate of water supply: 250 liter/capita/day. (08 Marks)

OR

- 8 a. Explain the zeolite process of water softening with sketch. Write down the relevant chemical equation. (10 Marks)
 b. Explain with a line diagram the “Nalagonda Technique” of defluridation. (06 Marks)
 c. Explain reverse osmosis process of softening of water. (04 Marks)

Module-5

- 9 Briefly explain the following with the help of neat sketch.
 a. Economical diameter of raising main
 b. Sluice valve
 c. Air valve
 d. Fire hydrant. (20 Marks)

OR

- 10 a. Discuss the various methods of distribution system. (06 Marks)
 b. With sketches, explain briefly dead end system and grid iron system of distribution networks. (08 Marks)
 c. Briefly explain the surface and elevated reservoirs. (06 Marks)

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Fifth Semester B.E. Degree Examination, July/August 2022 Construction Management and Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

Note : 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of Normal Distribution function table is permitted.

Module-1

- 1 a. What are the functions of Management? Explain any four of them. (08 Marks)
b. The activity data of a project is given in the table below :

Activity	Preceding Activity	Duration (Days)
A	-	10
B	-	9
C	A	9
D	A	8
E	B	7
F	B	11
G	D, E	5

Draw the Network diagram, identify and critical path and project duration using CPM.

(08 Marks)

- c. Discuss the characteristics of Autocratic Management Styles. (04 Marks)

OR

- 2 a. Below given table pertains to the list of activities and their time estimates of a Job.

Activity	Optimistic Time (Days)	Most likely time (Days)	Pessimistic Time (Days)
1 – 2	3	7	10
1 – 3	4	8	13
2 – 4	2	2	7
3 – 4	5	8	10

- i) Draw the Network and determine the critical path.
ii) What is the expected completion time with the probability of 85% (Take Probability factor (Z) = 1.038)? (10 Marks)
b. Discuss on types of Project plans. (06 Marks)
c. Explain the Gantt chart with sketch. (04 Marks)

Module-2

- 3 a. What are the factors affecting the Labour output? (08 Marks)
b. Estimate the hourly production in bulk volume (LCM) of a backhoe with bucket capacity of 1.14m³ that is employed on excavation of a foundation, which is 3m deep in hard digging soil. The excavated earth is to be loaded in waiting dump trucks, placed at a swing angle of 75°. The expected performance efficiency is 81% with the following data.
i) Ideal output of loose soil in M³ (LCM) = 180 LCM.
ii) Equipment conversion factor operating at optimum depth = 0.80.
iii) Correction factors are : Soil factor per hard digging = 0.67 ;
Load factor for loading into vehicle = 0.80 ; Swing factor = 1.05. (08 Marks)
c. Discuss on Class of Labour. (04 Marks)

OR

- 4 a. Enumerate the classification of Construction equipment with sub class. (10 Marks)
 b. What are the functions of Material Management? (05 Marks)
 c. A Company has purchased an equipment whose first cost is Rs 1,50,000 with an estimated life of eight years. The estimated salvage value of the equipment at the end of life time is Rs 25,000. Determine the depreciation and the book value for period 4 using the Straight Line method of depreciation. (05 Marks)

Module-3

- 5 a. Define Quality and describe Quality Control and Quality Assurance in Construction. (08 Marks)
 b. Explain the Total Quality Management process in Construction. (08 Marks)
 c. Discuss on need of Engineering ethics. (04 Marks)

OR

- 6 a. What are the Safety procedures to be adopted during Excavation? (06 Marks)
 b. Explain Safety through Legislation , Safety campaign and Insurances. (08 Marks)
 c. Briefly write about : i) Gifts and bricks ii) Whistle blowing. (06 Marks)

Module-4

- 7 a. Explain the principles of Engineering Economics. (08 Marks)
 b. An engineer has two bids for an excavator to be installed in a new building. The details of the bids for the excavator are as follows :

Bid	Engineer's Estimates		
	Initial cost (Rupees)	Service Life (Years)	Annual operation and Maintenance cost (Rupees)
Company X	10, 50, 000	15	60,000
Company Y	11, 00, 000	15	70,500

Determine which bid should be accepted , based on the Present Worth method of comparison assuming 18%, Interest rate , Compounded annually. (12 Marks)

OR

- 8 a. Define the following terms related to Engineering Economics : i) Present worth ii) Future worth iii) Marginal cost iv) Opportunity cost v) Capitalized cost. (10 Marks)
 b. Determine the effective interest rate for a nominal annual rate of 6 percent that is compounded : i) Semi annually ii) Quarterly iii) Monthly iv) Daily. (10 Marks)

Module-5

- 9 a. Write briefly about four functions of an Entrepreneur. (04 Marks)
 b. Summarize the role of MSME in Economic development. (08 Marks)
 c. Discuss in detail about stages in Entrepreneurial process. (08 Marks)

OR

- 10 a. Write short note on Exporting. (04 Marks)
 b. Discuss the different types of Feasibility study. (08 Marks)
 c. Explain the scope and role of following agencies : i) KIADB ii) KSFC. (08 Marks)

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Fifth Semester B.E. Degree Examination, July/August 2022

Analysis of Indeterminate Structures

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 Analyse the continuous beam ABCD shown in Fig Q1 by slope deflection method and draw SFD and BMD.

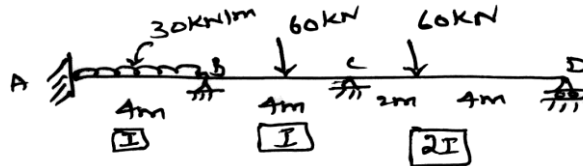


Fig Q1

(20 Marks)

OR

- 2 Analyse the frame shown in Fig Q2 by slope deflection method and draw BMD.

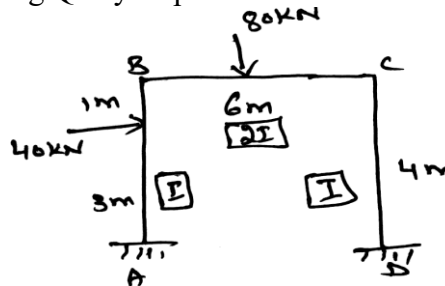


Fig Q2

(20 Marks)

Module-2

- 3 Analyse the continuous beam shown in Fig Q3 by moment distribution method and draw BMD.

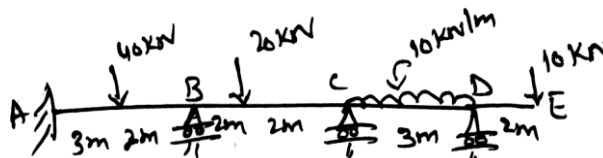


Fig Q3

(20 Marks)

OR

- 4 Analyse the frame shown in Fig Q4 by moment distribution method and draw BMD and SFD.

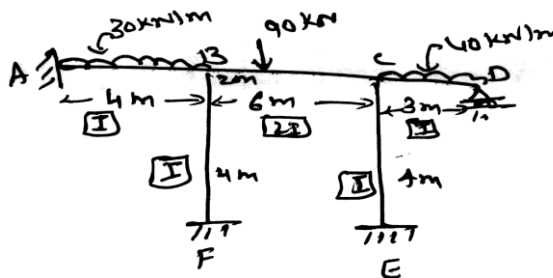


Fig Q4

(20 Marks)

Module-3

- 5 Analyse the beam shown in Kani's method in Fig Q5. Take $EI = 1$. Draw BMD.

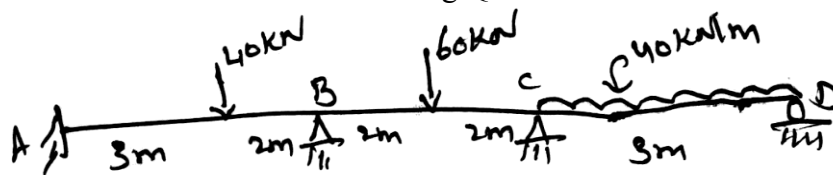


Fig Q 5

(20 Marks)

OR

- 6 Analyse the frame shown in By Kani's method in Fig Q6. Draw BMD.

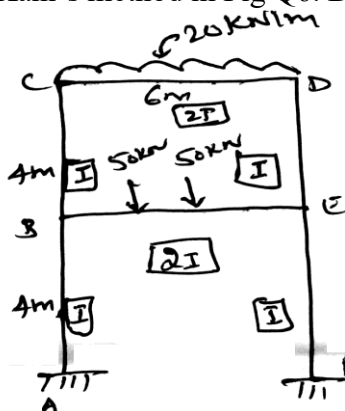


Fig Q 6

(20 Marks)

Module-4

- 7 Analyse the two spanned beam shown in Fig Q7, by flexibility matrix method. Draw BMD.

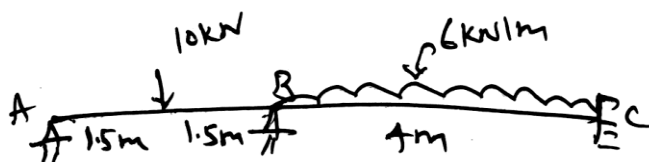


Fig Q7

(20 Marks)

OR

- 8 Analyse the frame shown in Fig Q8 by using flexibility matrix method.

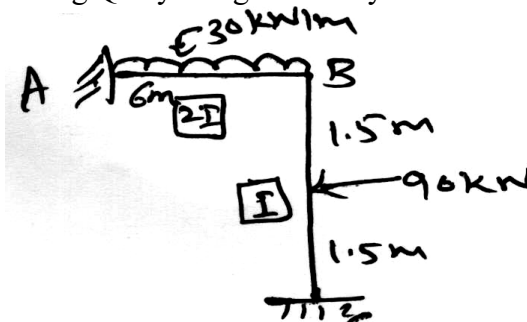


Fig Q8

(20 Marks)

Module-5

- 9 Analyse the portal frame shown in Fig Q9 by stiffness matrix method. Draw BMD.

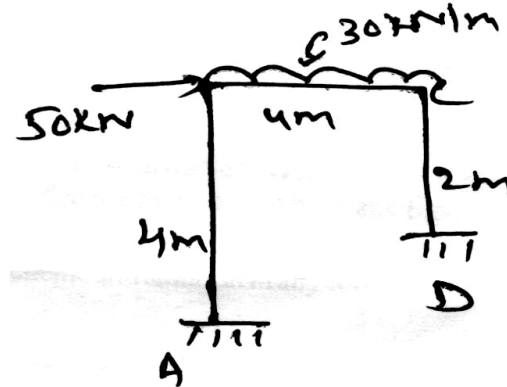


Fig Q9

(20 Marks)

OR

- 10 Analyse the pin jointed frame shown in Fig Q10, by stiffness matrix method. The area of member in sq mm shown in parentheses.

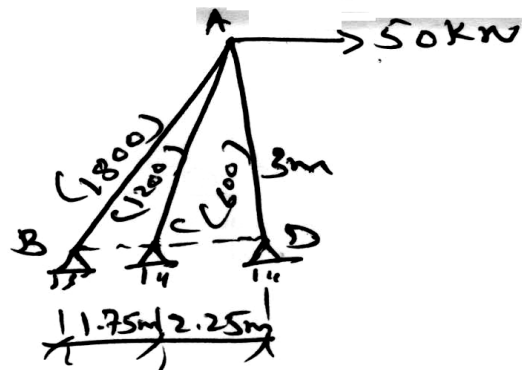


Fig Q10

(20 Marks)

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Fifth Semester B.E. Degree Examination, July/August 2022 Design of RC Structural Elements

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. IS456-200 and SP(16) is allowed.**

Module-1

- 1 a. Explain the philosophy and principals of Limit state method of design. (08 Marks)
- b. Write brief note on :
 - i) Partial safety factors for load and material
 - ii) Characteristics load
 - iii) Characteristics strength (06 Marks)
- c. Explain balanced, under reinforced and over-reinforced sections. (06 Marks)

OR

- 2 a. Differentiate between working stress method and limit state method of RCC designs. (06 Marks)
- b. A rectangular section beam 200mm wide by 450mm overall depth is reinforced with 3 bars of 16mm diameter at an effective depth of 420mm. Two hanger bars of 12mm diameter are provided at the expression face. The effective span of the beam is 5m. The beam supports a service load of 10kN/m. If $f_{ck} = 20\text{N/mm}^2$ and $f_y = 415\text{N/mm}^2$. Compute : i) The short term deflection ii) The long term deflection. (14 Marks)

Module-2

- 3 a. A singly reinforced beam 250mm × 500mm is reinforced with 4 bars of 16mm diameter with an effective cover of 50mm. Effective span of the beam is 6m. Assuming M₂₀ grade concrete and Fe415 steel determine central concentrated load 'P' that can be carried by the beam in addition to its self weight. (10 Marks)
- b. Determine the area of reinforcement required for a singly reinforced concrete section having breadth of 300mm and effective depth 600mm to resist a factored moment of 200kN-m. Take $f_{ck} = 20\text{N/mm}^2$ and $f_y = 415\text{ N/mm}^2$. (10 Marks)

OR

- 4 a. A RCC beam 300mm × 500mm is reinforced with 4 bars of 16mm diameter with an effective cover of 50mm. The beam is simply supported over a span of 5m. Find the maximum permissible udl on the beam. Use M₂₀ grade concrete and Fe500 steel. (10 Marks)
- b. Calculate the moment of resistance of RCC rectangular section 250mm wide and 500mm deep. If is reinforced with 6 bars of 20mm diameter on tension side and 3 bars of 20mm diameter on compression side. The effective cover for both tension and compression reinforcement is 40mm. Use M₂₀ grade concrete and Fe415 steel for the design. (10 Marks)

Module-3

- 5 Design a cantilever beam having 1.1m projection. The live load on the beam is 6kN/m use M₂₅ concrete and Fe415 steel for the design. (20 Marks)

OR

- 6 A T-beam slab floor has a slab 150mm thick spanning between T-beams which are spaced at 3m center to center. The beam is having clear span of 10m. The end bearings are 450mm thick. The live load on the floor is 4kN/m^2 . Using M_{20} concrete and Fe415 steel design one of the intermediate beam (20 Marks)

Module-4

- 7 Design a slab for a room size $7\text{m} \times 3\text{m}$ which carries a live load of 2.5kN/m^2 . Use M_{20} concrete and Fe 415 steel for the design. (20 Marks)

OR

- 8 Design a dog legged stair case for a room $2.5 \times 4.5\text{m}$. The floor to floor height is 3.2m. Take live load 3kN/m^2 . Assume bearing 230mm. Use M_{20} concrete and Fe415 steel for the design. (20 Marks)

Module-5

- 9 a. Determine reinforcement to be provided in a square column subjected to biaxial bending with following details.
 i) Size of the column = $450\text{mm} \times 450\text{mm}$
 ii) Concrete M_{25}
 iii) Steel : Fe415
 iv) Factored moment = 200kN-m
 v) Factored load = 2500kN
 If the arrangement of steel i) on two sides ii) on four sides. (12 Marks)
- b. Determine the reinforcement to be provided in a short column subjected to biaxial bending with the following data :
 i) Size of the column = $400\text{mm} \times 600\text{mm}$
 ii) Concrete mix = M_{25}
 iii) Characteristics strength of steel = 415N/mm^2
 iv) Factored load $P_u = 1600\text{kN}$
 v) Factored moment acting parallel to the longer direction $m_{ux} = 120\text{kN-m}$
 vi) Factored moment acting parallel to the dimension $M_{uy} = 90\text{kN-m}$. (08 Marks)

OR

- 10 Design a rectangular footing for a column $400 \times 600\text{mm}$ to carry a axial load of 1500kN . The safe bearing capacity of soil is 180kN/m^2 . Take M_{20} concrete and Fe415 steel for the design. (20 Marks)

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Fifth Semester B.E. Degree Examination, July/August 2022 Basic Geotechnical Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With the help of 3-phase diagram define: void ratio, porosity, water content and degree of saturation. (08 Marks)
- b. Derive from first principles, the following phase relation:

$$\gamma_d = \frac{(1 - n_a)G \cdot \gamma_w}{1 + wG}$$
(06 Marks)
- c. With the help of particle size distribution curve explain well graded soil, uniformly graded soil and gap graded soil. (06 Marks)

OR

- 2 a. Explain plasticity chart with a neat sketch and its use in classification of fine grained soil. (08 Marks)
- b. How many cubic meters of soil can be formed with a void ratio of 0.5 from 100 cubic meters of soil having void ratio of 0.7? (06 Marks)
- c. A soil sample consisting of particles size ranging from 0.5mm to 0.01mm is put on the surface of still water tank 5 mts deep. Calculate the time of settlement of the coarsest and finest particles of the sample to the bottom of the tank. Assume average specific gravity of soil particles as 2.66 and viscosity of water as 0.01 poise. (06 Marks)

Module-2

- 3 a. Explain with sketches the various soil structures. (06 Marks)
- b. Explain any two clay minerals with the help of neat sketches. (08 Marks)
- c. List and explain the factors affecting compaction. (06 Marks)

OR

- 4 a. What are the differences between standard and modified Proctor's compaction tests? (06 Marks)
- b. Explain Electrical diffused double layer and adsorbed water. (06 Marks)
- c. The observations of a standard Proctor's compaction test are as follows:

Dry density kN/m ³	16.16	17.06	18.61	18.95	18.78	17.13
Water content, %	5	8.81	11.25	13.05	14.40	19.25

Plot the compaction curve and determine OMC and $\gamma_{d \text{ max}}$. Also compute void ratio and degree of saturation at optimum condition. Take $G = 2.77$. (08 Marks)

Module-3

- 5 a. Define Darcy's law. Derive the relationship between discharge velocity and seepage velocity. (06 Marks)
- b. Derive an expression to obtain coefficient of permeability under falling head condition. (06 Marks)
- c. A clay structure of thickness 8m is located at a depth of 6m below the ground surface. It is overlaid by fine sand and the water table is located at a depth of 2m below ground surface. For fine sand submerged unit weight is 10.2kN/m³. The moist unit weight of sand located above water table is 16kN/m³. For clay layer $G = 2.76$ and $w = 25\%$. Compute the effective stress at the middle of clay layer. (08 Marks)

OR

- 6 a. What is a flownet? Briefly explain the characteristics and uses of flownets. (08 Marks)
 b. Explain the terms: Total stress, effective stress and neutral stress. (06 Marks)
 c. Calculate the coefficient of permeability of a soil sample, 6 cms in height and 50cm^2 in cross-sectional area, if a quantity of water equal to 430cc passed down in 10 minutes, under an effective constant head of 40cms. On oven drying the test specimen has a mass of 498gms. Taking the specific gravity of soil solids as 2.65, calculate the seepage velocity of water during the test. (06 Marks)

Module-4

- 7 a. Explain Mohr-Coulomb theory of shear strength. (06 Marks)
 b. Explain the advantages and disadvantages of direct shear test over triaxial shear test. (06 Marks)
 c. An unconfined compression test was conducted on an undisturbed sample of clay. The sample had a diameter of 38mm and was 80mm long. The load at failure was measured as 30N and the axial deformation of the sample at failure was 12mm. Determine the unconfined compressive strength and undrained shear strength of clay. (08 Marks)

OR

- 8 a. Explain sensitivity and thixotropy. (06 Marks)
 b. Explain Vane shear test with a neat sketch. (06 Marks)
 c. The triaxial tests carried out on soil samples gave the following results:

Confining Pressure., kN/m^2	50	100	150
Deviator stress, kN/m^2	76	132	186
Pore water Pressure., kN/m^2	35	59	83

Plot Mohr's circle and obtain effective shear parameters.

(08 Marks)

Module-5

- 9 a. Explain spring analogy theory of consolidation of soil. (06 Marks)
 b. What is preconsolidation pressure? How it is determined by Casagrande's graphical method? (06 Marks)
 c. An undisturbed sample of clay, 24mm thick, consolidated 50% in 20 minutes, when tested in laboratory with drainage allowed at top and bottom. The clay layer from which the sample was obtained is 4m thick in field. How much time will it take to consolidate 50% with double drainage? If the clay stratum has only single drainage, calculate the time to consolidate 50%. Assume uniform distribution of consolidation pressure. (08 Marks)

OR

- 10 a. Explain pre consolidated, normally consolidated and underconsolidated soil. (06 Marks)
 b. With a neat sketch, explain the square root of time fitting method of determining coefficient of consolidation. (06 Marks)
 c. In a consolidation test the void ratio of soil sample decreases from 1.20 to 1.10, when the pressure is increased from 160 to 320kN/m^2 . Determine coefficient of consolidation if the coefficient of permeability is $8 \times 10^{-7}\text{mm/s}$. (08 Marks)

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CBCS SCHEME

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18CV55

Fifth Semester B.E. Degree Examination, July/August 2022 Municipal Wastewater Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the Need of Sanitation. (04 Marks)
b. Explain the method of sewage disposal with advantages and disadvantages. (10 Marks)
c. Explain the principles of good house drainage. (06 Marks)

OR

- 2 a. Explain the time of concentration with its components. (08 Marks)
b. Assuming that the surface on which the rainfalls in a district is classified as follows :
20% of the area consists of roof for which the runoff ratio is 0.9, 20% of the area consists of pavements for which the runoff ratio is 0.85, 5% of the area consists of paved yards of houses for which runoff is 0.80, 15% of area consists of macadam roads for which run off ratio is 0.40, 35% of the area consists of lawns, gardens and vegetable plants for which the runoff ratio is 0.10 and the remaining 5% of the area is wooded for which the runoff ratio is 0.05, determine the co-efficient of runoff for the area. If the total area of the district is 36 hectares and the maximum rain intensity is taken as 5cm/hr, what is the total run off for the district? (08 Marks)
c. What is wet weather flow? Discuss the factors affecting wet weather flow. (04 Marks)

Module-2

- 3 a. Explain briefly :
i) Self cleaning velocity
ii) Non-scouring velocity. (08 Marks)
b. State hydraulic elements for circular sewer for full flow and half flow conditions. (12 Marks)

OR

- 4 a. With neat sketch flow diagram, explain unit operation and process of Municipal Waste Water Treatment. (06 Marks)
b. The BOD₅ of a waste water is 150mg/l at 20°C, the 'K' value is known to be 0.23 per day, what would BOD₈ be if the test was run at 15°C? (08 Marks)
c. Explain the physical, chemical and biological characteristics of waste water. (06 Marks)

Module-3

- 5 a. Discuss briefly with neat sketch Grit Chamber, oil and grease removal tank, screen's and its types. (12 Marks)
b. Explain self purification of natural water bodies; also explain various factors affecting self purification. (08 Marks)

OR

- 6 a. Discuss zones of purification with neat sketch. (08 Marks)
 b. A city discharges 1500 litres per second of sewage into a stream whose minimum rate of flow is 6000 liters per second. The temperature of sewage as well as water is 20°C . The 5 day B.O.D at 20°C for sewage is 200mg/l and that of river water is 1 mg/lit. The D.O. content of sewage is zero and that of the stream is 90% of the saturation D.O. If the minimum to be maintained in the stream is 4.5mg/ l, find out the degree of sewage treatment required. Assume the de-oxygenation coefficient as 0.1 and re-oxygenation co-efficient as 0.3. (12 Marks)

Module-4

- 7 a. With a neat diagram, explain activated sludge process. (08 Marks)
 b. Explain with neat sketch the working of trickling filter. What is the principle on which it is working? (08 Marks)
 c. Explain rotating biological contactor. (04 Marks)

OR

- 8 a. Explain principle of stabilization ponds with a diagram. (08 Marks)
 b. Explain :
 i) Thickness
 ii) Drying Bed
 iii) Equalization. (12 Marks)

Module-5

- 9 a. Discuss in briefly Nitrification and Denitrification process. (04 Marks)
 b. Explain with neat sketch electro coagulation method. (08 Marks)
 c. What do you mean by low cost treatment and explain various methods involved in cost treatment. (04 Marks)

OR

- 10 a. Discuss in briefly :
 i) Eco – toilet
 ii) Soak pits. (10 Marks)
 b. Design the dimensions of a septic, tank for a small colony of 150 persons provided with an assured water supply from the municipal head works at a rate of 120 liters per person per day. Assume any data, you may need. (10 Marks)

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Fifth Semester B.E. Degree Examination, July/August 2022 Highway Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Briefly explain the following:
 - i) Characteristic of road transport.
 - ii) Indian Road Congress.
 - iii) Central Road Research Institute. (10 Marks)
- b. Briefly explain the salient features of 3rd and 4th 20 year road development plans in India. (10 Marks)

OR

- 2 a. List the factors affecting the alignment. Briefly explain the engineering surveys for highway alignment. (10 Marks)
- b. Determine the length of different categories of roads in a state in India by the year 2001, using the third road development formula and the following data:

Total area of the state = 80,000 sq.km
Total number of towns as per 1981 census = 86
Overall road density aimed at = 82 km/100 sq. km area. (10 Marks)

Module-2

- 3 a. Define friction. List the factors affecting friction or skid resistance. (05 Marks)
- b. Define Camber. List and write neat sketches of different shapes of Camber. (05 Marks)
- c. Define superelevation. Derive an expression for superelevation with neat sketch. (10 Marks)

OR

- 4 a. Define transition curve and list the functions of transition curves in the horizontal alignment of highway. (05 Marks)
- b. Calculate the minimum sight distance required to avoid a head-on collision of two cars approaching from the opposite directions at 90 and 60kmph. Assume a reaction time of 2.5 seconds, coefficient of friction is 0.7 and a brake efficiency of 50 percent in both the cases. (07 Marks)
- c. The speeds of overtaking and over taken vehicles are 70 and 40kmph respectively on a two way traffic road. The average acceleration during overtaking may be assumed as 0.99m/sec².
 - i) Calculate safe overtaking sight distance.
 - ii) What is the minimum length of overtaking zone?
 - iii) Draw a neat sketch of the overtaking zone and show the positions of the sign posts. (08 Marks)

Module-3

- 5 a. Explain briefly the desirable properties of subgrade soil. (06 Marks)
 b. List the various properties of road aggregates and the tests to be conducted to find each property of aggregate. (06 Marks)
 c. What are the various tests carried out on Bitumen? Briefly mention the uses of each test. (08 Marks)

OR

- 6 a. Briefly outline the advantages and limitations of rigid pavement. (04 Marks)
 b. Draw a sketch of flexible pavement cross section and show the component parts. Enumerate the functions of each component of the pavement. (08 Marks)
 c. Explain ESWL with sketch. Briefly explain the graphical method of determination of ESWL for dual wheel assembly. (08 Marks)

Module-4

- 7 a. With a neat sketch, explain the method of determining the aggregate-bituminous mixes proportioning by Rothfuch's method. (10 Marks)
 b. What do you understand by wet-mix macadam? Briefly explain construction steps and quality control checks during the construction of WMM base course. (10 Marks)

OR

- 8 a. Explain the various steps in the construction of Dense Bituminous Macadam pavement. (10 Marks)
 b. Explain step by step, the procedure of Dry Lean Concrete sub base course. (10 Marks)

Module-5

- 9 a. What are the requirements of Highway Drainage system? (05 Marks)
 b. Briefly explain with neat sketches:
 i) Lowering of water table
 ii) Control of seepage flow. (10 Marks)
 c. Briefly explain the procedure for design of filter material. (05 Marks)

OR

- 10 a. Calculate the annual cost of a stretch of highway from the following particulars:

Item	Total cost Rs. in lakhs	Estimated life, years	Rate of interest, %
Land	35	100	6
Earth work	40	40	8
Bridges, culvert and drainage	50	60	8
Pavement	100	15	10
Traffic signs and real appurtenances	15	5	10

- The average cost of maintenance of the road is Rs.1.5 lakhs per year. (10 Marks)
 b. List the factors to be considered for evaluation of vehicle operation cost. (05 Marks)
 c. List the advantages of implementing road projects through public private partnership modes. (05 Marks)

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CBCS SCHEME

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Question Paper Version : A

Fifth Semester B.E. Degree Examination, July/August 2022

Environmental Studies

(COMMON TO ALL BRANCHES)

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the hundred questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. The word Tsunami is derived from Two Japanese word
 a) TSU (Big) and Nami (flow) b) TSU(harbor) and Name(wave)
 c) TSU (Big wave)and Nami (wave) d) None of the above.

 2. Green house gasses are
 a) Chloro fluoro carbon b) Oxygen
 c) Chlorine d) Chloro Benzene.

 3. Taj Mahal at Agra may be damaged by
 a) Sulphur Dioxide b) Chlorine c) Earth quake d) All of these.

 4. Ozone day is observed on
 a) January 30 b) April 21 c) September 16 d) December 25.

 5. ELISA Test is used to detect
 a) Malaria b) AIDS c) Cholera d) Tuberculosis.

 6. Earth day is observed on
 a) 1st December b) 5th June c) April 22nd d) 1st January.

7. Lead Poisoning may cause
 - a) Reduction in hemoglobin
 - b) Kidney damage
 - c) Mental retardation
 - d) All of these.
8. Noise pollution limits at residential areas at Night is
 - a) 45dB
 - b) 80 dB
 - c) 55dB
 - d) 90dB.
9. Major purpose of most of the dams around the world is
 - a) Power generation
 - b) Drinking water supply
 - c) Flood control
 - d) Irrigation.
10. Herpetology is a branch of science which deals with
 - a) Reptiles only
 - b) Amphibians and Reptiles
 - c) Mammals
 - d) Fishes.
11. Water logging is a phenomenon in which
 - a) Crop patterns are rotated
 - b) Soil root zone becomes saturated due to over irrigation
 - c) Erosion of soil occurs
 - d) All of the above.
12. Coral reefs in India can be seen in
 - a) Goa
 - b) Himalayan region
 - c) Andaman and Nicobar Islands
 - d) Uttar Pradesh.
13. In which year salient valley was declared as National park
 - a) 1984
 - b) 1987
 - c) 1983
 - d) 1989.
14. SMOG is a mixture of
 - a) Snow and fog
 - b) Smoke and fog
 - c) Sulphur dioxide and fog
 - d) Snow and dust
15. Effect of modern Agriculture on soil is due to
 - a) Erosion
 - b) Acidification
 - c) Salinization
 - d) All of the above.
16. What percentage of its geographical area of a country should be under forest cover?
 - a) 23%
 - b) 43%
 - c) 13%
 - d) 33%.
17. About _____% of the earth surface is covered by water
 - a) 53%
 - b) 19%
 - c) 71%
 - d) 90%.
18. Forest prevent soil erosion by binding soil particles in their
 - a) Stems
 - b) Roots
 - c) Leaves
 - d) Buds.
19. Blue baby syndrome (methane moglabinemia) is caused by contamination of water due to
 - a) Phosphates
 - b) Sulphur
 - c) Arsenic
 - d) Nitrates.

20. India has the largest share of which of the following
 a) Manganese b) Mica c) Copper d) Diamond.
21. What is the minimum Allowable concentration of fluorides in drinking water
 a) 1.0 mg/L b) 1.25 mg/L c) 1.50 mg/L d) 1.75 mg/L.
22. Nitrogen fixing Bacteria exists in – of plants
 a) Leaf b) Roots c) Stem d) Flower.
23. ____ are referred to as Earth's lungs
 a) Forests b) Carbon cycle c) Water sources d) Mines.
24. GIS means
 a) Geographic information source
 b) Spatial information system
 c) Geological information system
 d) Geographical information system.
25. Cholera and Typhoid are caused by
 a) Worms b) Virus c) Bacteria d) Fungus.
26. Remote sensing includes gathering of
 a) Images b) Changes c) Movements d) Sounds.
27. First satellites launched by India was and when
 a) Aryabhata - 1975 b) Apple - 1981 c) Bhaskara - 1981 d) Kalpana – 1983.
28. Biomedical waste may be disposed off by
 a) Incineration b) Autoclaving and land filling
 c) Both a) and b) d) None of the above.
29. Which atmospheric layer is closest to the earth surface
 a) Mesosphere b) Troposphere c) Stratosphere d) Thermosphere.
30. Sulabha Bio-gas plants are based on the use of
 a) Human excreta b) Cattle dung c) Agricultural wastes d) All of these.
31. National law day is celebrated on
 a) 1st April b) 26th January c) 26th November d) 1st June.
32. Geology is the study of
 a) Animals b) Mammals c) Science of earth d) Space.
33. Housing has become inaccessible to the poor due to
 a) Increased population b) Non-availability of food
 c) High cost d) None of above.

34. Major source of fluoride available in
 a) River water b) Ground water c) Food product d) Both a) and c).
35. World AIDS Day is celebrated on
 a) 15th Aug b) 1st Sept c) 1st Jan d) 1st December.
36. A food web consists of
 a) Portion of food chain b) Producers, Consumers and Decomposers
 c) Inter locking of food chains d) A set of similar consumers.
37. Eutrofication is
 a) An improved water quality status of lakes
 b) The Result of Accumulation of plant nutrients in water bodies
 c) A process in carbon cycle
 d) A water purification techniques.
38. EIA can be expanded as
 a) Environmental and Industrial Activities
 b) Environment impact activities
 c) Environment impact assessment
 d) Environmental internal activities.
39. Identity the non renewable source of energy from the following :
 a) Coal b) Tidal power c) Wind power d) Wave power.
40. An Alternative eco-friendly fuel for Automobiles is
 a) Petrol b) Diesel c) CNG d) Kerosene.
41. Chernobyl nuclear disaster occurred in the year
 a) 1984 b) 1952 c) 1986 d) 1987.
42. Minamata disease is caused by
 a) Lead b) Mercury c) Cadmium d) Arsenic.
43. Bhopal gas Tragedy occur due to leakage of
 a) Methyl/iso cyanate b) Sulphur dioxide c) Mustard gas d) Methane gas.
44. Nuclear fusion reaction occurs in the
 a) Sun b) Stars c) Hydrogen bomb d) All of these.
45. Definition of noise is
 a) Loud sound b) Unwanted sound
 c) Constant sound d) Sound of high frequency.
46. Demography is the study of
 a) Animal behavior b) Geography c) Rivers d) Population growth.

47. Acid Rain is caused by increase in the atmospheric concentration of
 a) Ozone and dust b) SO₂ and NO₂ c) SO₃ and CO d) CO₂ and CO.
48. Global warming could affect
 a) Climate b) increase in sea level
 c) Melting of glaciers d) All of these.
49. The Wild Life Protection Act in India was passed in
 a) 1978 b) 1972 c) 1986 d) 1992.
50. Environment Protection Act was enacted in year.
 a) 1986 b) 1974 c) 1992 d) 1984.
51. Environment protection is the fundamental duties of the citizen of India under the article
 a) 48 – A b) 47 – A c) 51 – A d) 21 – B.
52. The Forest Conservation Act was enacted in the year
 a) 1986 b) 1974 c) 1980 d) 1972.
53. The major contributors of the acid rain are known as
 a) Precursors b) Processors c) Protons d) Pollutants.
54. Typical Acid Rain pH is
 a) 4.0 b) 5.5 c) 6.0 d) 4.0 - 6.0.
55. Pesticide causes
 a) Eye irritation b) Skin irritation
 c) Respiratory ailments d) All of Above.
56. Percentage of fresh water available below the earth is
 a) 2.8% b) 2.2% c) 9.8% d) 2.15%.
57. The Quantity of solar energy received by the earth
 a) 71% b) 99% c) 45% d) 15%.
58. Which pyramid is always upright
 a) Energy b) Biomass Number d) Food chain.
59. The leader of Chipko movement is
 a) Sunderlal Bahuguna b) Medha Patkar
 c) Vandana Shiva d) Suresh Heblikar.
60. Padma Shri was received by immense contribution toward conservation of Trees Recently
 a) Saalumarada Thimakka b) Tulsi Gowda
 c) Vandana d) Ramnath.
61. The first international earth summit was held in
 a) New Delhi b) Kyoto c) Stockholm d) Rio de Janeiro.

62. Ozone layer thickness is measured in
a) PPM b) PPb c) Decibels d) Dobson unit.
63. Who first discovered ozone hole
a) Shankliln b) Charles Fabry c) Henri Baisson d) John Macculay.
64. Noise is measured in
a) Decibels b) Jouls c) PPM d) NTU.
65. Terrace farming is practiced in
a) Coastal areas b) Deserts c) plains d) Hills.
66. _____ was awarded Padma Shri for constructing primary school at his village by his earnings
a) Harekala Hajabba b) Tulsi gowda c) Chalus Fabry d) None of above.
67. Geothermal energy is a
a) Heat energy b) Current energy c) Wind energy d) Solar energy.
68. Cauvery water dispute is between
a) Karnataka and Maharastra b) Karnataka and Kerala
c) Karnataka and Tamil Nadu d) Uttar Pradesh and Madya Pradesh.
69. The word ecology is produced by
a) Ernst Haeckel b) Helena Curtis
c) Charels south wick d) Charles Alton.
70. Mining means
a) To conserve and preserve minerals b) To extract minerals and ores
c) To check pollution due to mineral resources d) None.
71. Fossil fuels are converted into energy by
a) Burning b) Cooling c) Sublimation d) Melting.
72. Physical pollution of water due to
a) Dissolved oxygen b) P^H c) Turbidity d) None of above.
73. Molasses from sugar industry is used to generate
a) Biodiesel b) Hydrogen c) Bio-ethanol d) Bio-methanol.
74. The most important fuel used by Nuclear power plant is
a) U – 235 b) U – 238 c) U – 245 d) U – 248.
75. Liquid waste generated from Bath rooms and kitchens are called
a) Domestic sewage b) Runoff c) Salvage d) All of above.

76. Water day is celebrated on this day
 a) 22nd March b) 25th April c) 15th June d) 5th June.
77. Loss of water content through the plants into the atmosphere is called
 a) Evaporation b) Vaporization c) Hydraulic cycle d) Transpiration.
78. The hydrological cycle is related to
 a) Water and electricity b) Water characterization
 c) Hydro power d) Water cycle and Balance.
79. The World Environment Day is celebrated on
 a) June 5th b) November 5th c) April 5th d) December 26th.
80. Nuclear power plant in Karnataka is located at
 a) Bhadravathi b) Sandur c) Raichur d) Kaiga.
81. Cell phones emits _____ type of Radiation
 a) Radio frequency Radiation b) UV radiation
 c) X ray radiation d) None of Above.
82. Water bearing rock readily Transmits water to well and springs
 a) Aquifer b) Porosity c) Permeability d) None of Above.
83. Karnataka State Pollution Control Board (KSPCB) was established in the year.
 a) 1947 b) 1982 c) 1986 d) 1974.
84. In the world's population, India Accounts for
 a) 10% b) 5% c) 17.7% d) 16%.
85. Which of the following is not a biodegradable pollutant
 a) Plastic b) Skins of vegetable and Fruits
 c) Dry leaves d) Paper.
86. Sundarbans is the name of place in
 a) Assam b) West Bengal c) Karnataka d) None of these.
87. Western ghats are locate in
 a) North East India b) Maharashtra c) Peninsular India d) Gujarat.
88. DDT is a
 a) Fungicide b) Pesticide c) Fertilizer d) Disinfectant.
89. Kala Azar is spread by
 a) Sand files b) Mosquitoes c) Rats d) Tapeworms.

90. SARA refer to
a) Severe acute respiratory syndrome
b) Self acute respiratory system
c) Severe acute respiratory system
d) Self accurate rest syndrome
91. Landslides are caused by
a) Earth quakes b) Dam buildings c) Mining d) All of these.
92. The virus that causes AIDS is
a) HIV b) TMV c) HCV d) None of these.
93. Rearing fish is called
a) Sericulture b) Fish culture c) Pisciculture d) Horticulture.
94. Percentage methane content of Biogas is
a) 50 to 70% b) 85% c) 5% d) 95%
95. Environment means
a) A beautiful landscape b) Industrial production
c) Air and water d) Sum of total all condition.
96. Taungya system is
a) Agro forestry b) Mining c) Exhaustible d) None of these.
97. Lithosphere means
a) Air b) Water c) Micro-organisms d) Rock and Soils.
98. Which among the following is a climate factor
a) Pressure b) Humidity c) Temperature d) All of these.
99. The forest is
a) Abiotic b) Biotic c) Both A and B d) None of these.
100. Store house of minerals is
a) Soil b) Water c) Forest d) All of these.

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CBCS SCHEME

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18CV61

Sixth Semester B.E. Degree Examination, July/August 2022 Design of Steel Structural Elements

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of IS 800-2007, steel table is permitted.

Module-1

- 1 a. What are the advantages and disadvantages of steel structures? (10 Marks)
- b. Distinguish between working stress design and limit state design of steel structures. (10 Marks)

OR

- 2 a. Calculate the shape factor of triangle. (10 Marks)
- b. Calculate 'M_p' for the continuous beam shown in Fig.Q2(b). Take load factor 1.5.

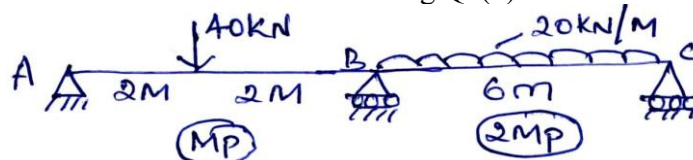


Fig.Q2(b)

(10 Marks)

Module-2

- 3 a. Explain the failure modes of bolted connection. (10 Marks)
- b. Design a bolted connection for a lap joint of plate thickness 16 mm and 12 mm to carry a factored load of 160 kN. Use M₁₆ and 4.6 grade bolts. (10 Marks)

OR

- 4 a. What are the advantages and disadvantages of welded connection? (10 Marks)
- b. A tie member of Truss consisting of angle section ISA 65 × 65 × 6 mm of Fe 410 grade is welded to 8 mm gusset plate. Design a weld to transmit a factored load of 150 kN. (10 Marks)

Module-3

- 5 a. Explain the failure modes of axial loaded column. (10 Marks)
- b. Determine the design compressive strength of ISHB300@576.8 N/m, Length of column is 3.5 m and both ends are pinned. (10 Marks)

OR

- 6 Design a single angle discontinuous strut to carry a factored load of 65 kN. The length of strut is 3m, between inter section. It is connected to 12 mm thick gusset plate by 20 mm diameter, 4.6 grade bolts. (20 Marks)

Module-4

- 7 a. Explain the factors effecting strength of tension members. (10 Marks)
- b. Design a tension member to carry factored load of 400 kN connected to shorter leg back to back. Length of member is 3m. (10 Marks)

OR

- 8 a. Explain Lug angles and column splices. (10 Marks)
b. Design slab base for a column made of ISHB250@536 N/m to carry axial working load of 520 kN. The grade of concrete is M₂₀ and grade of steel Fe 410. (10 Marks)

Module-5

- 9 a. Explain the factors effecting lateral stability of beams. (10 Marks)
b. Calculate the load carrying capacity of laterally restrained simply supported beam with ISMB500@86.9 kg/m section for an effective span of 5m. (10 Marks)

OR

- 10 Design a steel beam section for supporting hall for the following data:
Clear span = 6.5 m
End bearing = 200 mm
c/c spacing of beams = 3 m
Live load on beams = 12 kN/m²
Dead load = 3 kN/m²

(20 Marks)

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Sixth Semester B.E. Degree Examination, July/August 2022 Applied Geotechnical Engineering

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Missing data, if any may be suitably assumed.**

Module-1

- 1 a. What is sub-surface exploration? What are the objectives of soil exploration? (07 Marks)
- b. With a neat sketch explain seismic refraction method. (07 Marks)
- c. Determine the area ratios for the following soil sampler and comment the nature of samples obtained in each samplers
 - i) Core cutter 185mm OD 135mm ID
 - ii) Split Barrel 51mm OD 45mm ID
 - iii) Shellby tube 51mm OD 49mm ID

Which one you recommended to be used for getting good qualifier samples? (06 Marks)

OR

- 2 a. List the methods of dewatering technue used in the field and explain vacuum method. (06 Marks)
- b. List and explain different types of samples of soil. (06 Marks)
- c. Estimate the position of ground water table with following data by Hvorselve's method.
Depth upto which water is bailed out = 10.5m.

Water rise in 1 st day	=	0.63m
2 nd day	=	0.57
3 rd day	=	0.51m

(08 Marks)

Module-2

- 3 a. Distinguish between Boussinesq's theory with Westergaurd's theory of stress distribution. (06 Marks)
- b. Explain contact pressure distribution of soil. (06 Marks)
- c. A circular area 6m in diameter carries uniformly distributed load of 10kN/m². Determine the vertical stress at a depth of 2m, 4m, and 8m. Plot the variation of vertical stress with depth. (08 Marks)

OR

- 4 a. Explain :
 - i) Pressure bulb
 - ii) Pressure distribution on Horizontal plane
 - iii) Pressure distribution on vertical plane. (06 Marks)
- b. What are the different types of settlements? Explain. (06 Marks)
- c. A Normally consolidated clay layer is 18m thick. Natural water content is 45%, saturated unit weight is 18kN/m³ specific gravity is 2.7 and liquid limit is 63%. The vertical stress increment at centre of clay layer due to foundation load is 9kPA. Ground water table is at the surface. Determine the settlement. (08 Marks)

Module-3

- 5 a. With a neat sketches, explain types of earth pressure. (06 Marks)
- b. Describe Rebhann's graphical method of determining the active earth pressure on retaining wall. (08 Marks)

- c. A Retaining wall 7.5m high retains cohesionless, horizontal backfill. The top 3m of fill has a unit weight of 18kN/m^3 and $\phi = 30^\circ$ and the rest has a unit weight of 24kN/m^3 and $\phi = 20^\circ$. Determine using Rankine's theory, the distribution of active earth pressure and total active earth thrust. (06 Marks)

OR

- 6 a. With neat sketches, explain different types of slope failures. (06 Marks)
 b. Explain Swedish circle method of stability analysis of slopes for $C - \phi$ soils. (06 Marks)
 c. An embankment is to be constructed with $C = 20\text{kN/m}^2$, $\phi = 20^\circ$, $\gamma = 18\text{kN/m}^3$, F.S = 1.25 and height is 10m. Estimate side slope required. Taylor's stability numbers are as follows below table. Also find the factor of safety, if the slope is 1V: 2H given $\phi = 20^\circ$.

Slope angle	90	75	60	45	30	20	10
Sn	0.182	0.134	0.097	0.062	0.025	0.005	0

(08 Marks)

Module-4

- 7 a. Explain the types of shear failures with neat sketches. (06 Marks)
 b. With the help of neat sketches, explain the effect of water table and eccentric loading on bearing capacity of soil. (08 Marks)
 c. A square footing is to be constructed on a deep deposit of sand at a depth of 0.9m to carry a design load of 300kN with a factor of safety of 2.5. The ground water table may rise to the ground level during rainy season. Design the plan dimension of footing given $\gamma_{\text{sat}} = 20.8\text{ kN/m}^3$, $N_c = 25$, $N_q = 34$ and $N_\gamma = 32$. (06 Marks)

OR

- 8 a. List the assumptions and limitation made in Terzaghi's analysis. (06 Marks)
 b. With neat sketch, explain plate load test. (06 Marks)
 c. A square footing $2.8 \times 2.8\text{m}$ is built on a homogeneous bed of sand of density 18kN/m^3 and $\phi = 36^\circ$. If depth of foundation is 1.8m. Determine the safe load on footing. Take $F = 2.5$, $N_c = 27$, $N_q = 36$, $N_\gamma = 35$. (08 Marks)

Module-5

- 9 a. Explain the classification of piles based on the material and function. (08 Marks)
 b. Mention the situations where the pile foundation is necessary. (04 Marks)
 c. In a group of 16 pile diameter is 450mm and center to center spacing of the square group is 1.5m. If $C = 50\text{kN/m}^2$, determine whether the failure would occur with the pile acting individually, or as a group? Neglect bearing at the tip of the pile. All piles are 10m long. Take adhesion factor as 2 and Factor of safety 2.5. Also find safe allowable load. (08 Marks)

OR

- 10 Write a short notes on :
 i) Group capacity of piles (05 Marks)
 ii) Negative skin friction (05 Marks)
 iii) Under reamed piles (05 Marks)
 iv) Settlement of piles. (05 Marks)

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Sixth Semester B.E. Degree Examination, July/August 2022
Hydrology and Irrigation Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss the various processes involved in 'Hydrologic Cycle' using Horton's Engineering representation. (10 Marks)
- b. List the importance of hydrology with emphasis on global water availability. (10 Marks)

OR

- 2 a. Define Rain gauge. Describe with a neat sketch, the principle of working of Symon's non – recording gauge and its demerits. (06 Marks)
- b. What is Precipitation? Distinguish between Convection and Orographic precipitation. (07 Marks)
- c. Determine the optimum number of raingauges in a catchment area using the following data :
- Number of existing rain gauges = 08.
 - Mean annual rainfall at the gauges : 1000, 950 , 900 , 850 , 800 , 700 , 600 and 400mm.
 - Permissible error = 6%.
- (07 Marks)

Module-2

- 3 a. What is meant by 'Evaporation Losses'? Discuss the factors affecting evaporation. (08 Marks)
- b. Define 'Evapotranspiration'. Explain in brief the 'Lysimeter method' of estimating the same in the field. (06 Marks)
- c. What is the Evaporation, if 4.80 litres of water is removed from an evaporation pan of diameter 1.22m and the simultaneous rainfall measurement is 9.0mm? (06 Marks)

OR

- 4 a. Discuss the factors that affect infiltration. Explain with a neat sketch, measurement of infiltration using double ring infiltrometer. (10 Marks)
- b. A 6 hour storm produced rainfall intensities of 7, 18, 25, 12, 10 and 3mm/hour is successive one hour intervals over a basin of 800km². The resulting run - off is observed to be $264 \times 10^5 \text{ m}^3$. Determine ϕ - index for the basin. (10 Marks)

Module-3

- 5 a. Define the following :
- Basin recharge
 - Direct run off
 - Drainage density
 - Form factor
 - Overland flow.
- (10 Marks)
- b. What is Runoff? List and explain factors affecting it. (10 Marks)

OR

- 6 a. How the hydrograph is affected by the following :
- Shape of the basin
 - Non – uniform aerial distribution of rainfall.
- (06 Marks)
- b. Define 'Unit hydrograph'. With the help of neat sketch, explain the various components of a flood hydrograph. (06 Marks)

- c. Given the ordinates of a 4 – hour unit hydrograph. Derive the ordinates of 12 – hour unit hydrograph for the same catchment. What is the peak value of discharge and the corresponding time interval observed in 4-h and 12-h unit hydrograph. (08 Marks)

Time (Hours)	0	4	8	12	16	20	24	28	32	36	40	44
Ordinates of 4-h UH cm^3/sec	0	20	80	130	150	130	90	52	27	15	05	0

Module-4

- 7 a. Define Irrigation. Discuss in brief the benefits and ill – effects of irrigation. (08 Marks)
 b. Distinguish between : Direct Irrigation and Storage Irrigation. (06 Marks)
 c. What is Bhandara Irrigation? List its advantages and disadvantages. (06 Marks)

OR

- 8 a. Define Duty and Delta. Derive the relation between them. (06 Marks)
 b. Define the following :
 i) Permanent wilting point ii) Field capacity. (06 Marks)
 c. After how many days water supply is required to ensure good yield, if :
 Field capacity of soil = 30% ; Permanent wilting point = 12% ;
 Density of soil = 1.4g/cc ; Effective depth of root zone = 80cm ;
 Daily consumptive use = 15mm. Readily available moisture is 85% of available moisture. (08 Marks)

Module-5

- 9 a. Write an explanatory note on Canal classification on the basis of its alignment. (06 Marks)
 b. Enumerate the basic differences between Lacey's and Kennedy's theory. (06 Marks)
 c. A channel section is to be designed for the following data :
 Discharge $Q = 5$ cumecs ; Silt factor = 1.0 ; Side slope = 0.5H = 1V.
 Also determine bed slope of the channel. Use Lacey's theory. (08 Marks)

OR

- 10 a. With a neat sketch, explain different zones of a storage reservoir. (10 Marks)
 b. With a neat sketch, explain step – by – step procedure of determining reservoir capacity for a specific yield using the mass – inflow curve. (10 Marks)

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Sixth Semester B.E. Degree Examination, July/August 2022

Ground Improvement Techniques

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Write a short note on formation of rock. (07 Marks)
- b. What is reclaimed soil? List out different types of reclamation material. (07 Marks)
- c. Write a short note on surface compaction. (06 Marks)

OR

- 2 a. What are the factors to be considered for Ground Improvement Method? (07 Marks)
- b. Write a short note on soil profile, with a sketch. (07 Marks)
- c. Write short note on natural offshore deposits. (06 Marks)

Module-2

- 3 a. Explain briefly different methods of dewatering system, with sketches. (10 Marks)
- b. What is filter and its requirements? (05 Marks)
- c. What are factors affecting electro kinetic stabilization of soil? (05 Marks)

OR

- 4 a. Define drain. List out different drains and explain briefly with neat sketches. (08 Marks)
- b. Explain Sand drain, with a neat sketch. (07 Marks)
- c. Mention various design steps involved in dewatering system. (05 Marks)

Module-3

- 5 a. Define Cement Stabilization. What are the effects of cement stabilization on shear strength of soil? (07 Marks)
- b. Explain soil stabilization using flyash. (07 Marks)
- c. List out factors controlling characteristics of lime concrete clay. (06 Marks)

OR

- 6 a. Explain mechanism of lime stabilization along with the chemical changes involved? (08 Marks)
- b. What are the soil requirements for stabilization using asphalt? (07 Marks)
- c. Explain Sandwich technique. (05 Marks)

Module-4

- 7 a. List out different method vibro-displacement. Explain any two provide figures. (08 Marks)
- b. Explain different installation techniques of stone columns. (06 Marks)
- c. List out Jet grouting methods. Explain with neat figure. (06 Marks)

OR

- 8 a. Explain blasting techniques. List out disadvantages of blasting techniques. (07 Marks)
- b. Explain vibrofloatation with a neat sketch. (08 Marks)
- c. List out different types of grouting and brief them. (05 Marks)

Module-5

- 9 a. List out different types of geosynthetics and explain any three with neat sketch. Mention their applications. (12 Marks)
b. What are the application of Geosynthetics? (08 Marks)

OR

- 10 Write a short note on any five:

- a. Soil nailing.
- b. Soil reinforcement.
- c. Micropiles.
- d. Shotcrete.
- e. Gabions.
- f. Cribwalls.

(20 Marks)

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Sixth Semester B.E. Degree Examination, July/August 2022 Railways, Harbours, Tunneling and Airports

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss the conventional method of route alignment survey. (08 Marks)
b. Discuss the significance of road, rail, water and air transport. (06 Marks)
c. Explain the following :
i) Coning of wheel with diagram
ii) Turnout with a neat sketch
iii) Negative super elevation with a neat sketch. (06 Marks)

OR

- 2 a. List the requirements and explain the types of rails. (07 Marks)
b. Explain the functions and requirements of ballast. (05 Marks)
c. What would be the permissible speed on the curve if on a 8°MG track, the average speed of different trains is 50kmph and allowable Cant deficiency is of that of maximum Cant deficiency. (08 Marks)

Module-2

- 3 a. Explain the maintenance of railway track. (06 Marks)
b. Discuss the stabilization methods of track on poor soil. (06 Marks)
c. Demonstrate the modern methods of track maintenance. (08 Marks)

OR

- 4 a. Describe a sump yard with neat sketch. List the methods of stopping the rolling down wagons. (07 Marks)
b. Explain the classification of railway station. (06 Marks)
c. Mention the passenger amenities to be provided in railway station. (07 Marks)

Module-3

- 5 a. List the types of break water and discuss the characteristics of mound break water. (08 Marks)
b. Write a note on tunnel drainage and tunnel lining. (06 Marks)
c. With a neat sketch explain the linear plate method. (06 Marks)

OR

- 6 a. List the classification of harbours and draw a neat sketch of the layout of an artificial harbor with components. (10 Marks)
b. Explain the three systems of mechanical ventilation of tunnels. (10 Marks)

Module-4

- 7 a. Draw a neat sketch of an airport with open concept of runways and explain the functions of the component. (10 Marks)
b. List the factors to be considered while selecting a suitable site for a major airport and explain the features of preferential runway with sketch. (10 Marks)

OR

- 8 a. Write a note on parking and circulation area. (07 Marks)
 b. Explain the characteristics of air transport. (06 Marks)
 c. Explain the aircraft characteristics which affect the airport design. (07 Marks)

Module-5

- 9 a. Explain the procedure of determining the best direction of orienting the runway as per type-I wind rose diagram with assumed data. (10 Marks)
 b. A runway length required for landing at sea level in standard atmospheric condition is 3000m. Runway length required for take-off at a sea level site in standard atmospheric condition is 2500m. Aerodrome reference temperature is 25°C and that of standard atmosphere at aerodrome elevation of 150m is 14.025°C if the effective gradient is 0.5%. Determine the runway length is to be provided. (10 Marks)

OR

- 10 a. Explain the passenger facilities and services available at airport. (05 Marks)
 b. Explain the different types of lighting used in airport. (05 Marks)
 c. Wind rose data shown in table the permissible cross wind component is 25kmph. Determine the period the best direction of runway and total wind coverage.

Wind direction	Deviation of wind percentage		
	6.4 – 25 kmph	25 – 40 kmph	40 – 60 kmph
N	7.4	2.7	0.2
NNE	5.7	2.1	0.3
NE	2.4	0.9	0.6
ENE	1.2	0.4	0.2
E	0.8	0.2	0
ESE	0.3	0.1	0
SE	4.3	2.8	0
SSE	5.5	3.2	0
S	9.7	4.6	0
SSW	6.3	3.2	0.5
SW	3.6	1.8	0.3
WSW	1	0.5	0.1
W	0.4	0.1	0
WNW	0.2	0.1	0
NW	5.3	1.9	0
NNW	4	1.3	0.3

(10 Marks)

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18CV654

Sixth Semester B.E. Degree Examination, July/August 2022 Sustainability Concepts in Civil Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define sustainability and discuss the need for sustainable developments. (06 Marks)
b. Explain:
(i) Water Control and Prevention Act
(ii) Air Control and Prevention Act (14 Marks)

OR

- 2 a. Explain the three-pillar model of sustainability. (08 Marks)
b. What are the challenges to attain sustainable developments? (06 Marks)
c. What is MEA? List the key features. (06 Marks)

Module-2

- 3 a. Explain briefly the concept of Local Environmental issues. (10 Marks)
b. Write short notes on:
(i) Carbon foot print
(ii) Carbon Trading (10 Marks)

OR

- 4 a. Write a short note on Life Cycle Analysis. (10 Marks)
b. Explain:
(i) Global Warming
(ii) Ozone Layer Depletion (10 Marks)

Module-3

- 5 a. Define Green Building. Discuss any two certification programmes implemented in India for Green Building. (12 Marks)
b. Write a short note on Passive Solar Building Design. (08 Marks)

OR

- 6 a. How energy efficiency can be achieved in a sustainable building design? Explain. (10 Marks)
b. Explain briefly:
(i) Thermal storage (05 Marks)
(ii) Sustainable cities and sustainable transport (05 Marks)

Module-4

- 7 a. Define Energy. Explain various forms of energy. (10 Marks)
b. Write short notes on:
(i) Geothermal Energy (05 Marks)
(ii) Wind Energy (05 Marks)

OR

- 8** a. Briefly explain the methods of extraction of energy from oceans. (10 Marks)
b. Solar energy as a renewable form of energy. Explain briefly. (10 Marks)

Module-5

- 9** a. Write a short note on sustainable urbanization. (10 Marks)
b. Explain briefly the concept of sustainable Poverty Reduction. (10 Marks)

OR

- 10** a. Briefly explain the Social and Technological change for Sustainable Development. (10 Marks)
b. Write short notes on:
(i) Industrial Ecology (05 Marks)
(ii) Industrial Symbiosis (05 Marks)

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Seventh Semester B.E. Degree Examination, July/August 2022 Design of RCC & Steel Structures

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any *TWO* full questions, choosing *ONE* full question from each module.
2. Use of IS456, IS-800, SP(6) and steel tables are permitted.

Module-1

- 1 Design a combined footing slab beam type for two RCC column A and B separated by a distance of 4 m C/C. Column A is 500×500 mm and carries a load of 1250 kN and column B is 600×600 mm and carries a load of 1600 kN. Take SBC of soil is 200 kN/m^2 . Use M20 concrete and Fe415 steel. Draw the sectional elevation. (50 Marks)

OR

- 2 Design a Cantilever retaining wall to retain earth embankment 5 m high above ground level. The density of earth is 18 kN/m^3 and its angle of repose is 30° . The embankment is horizontal at its top. The SBC of may be taken as 200 kN/m^2 and the co-efficient of friction between soil and concrete is 0.5. Adopt M20 grade concrete and Fe415 steel. Also draw cross sectional elevation showing reinforcement details. (50 Marks)

Module-2

- 3 A line diagram of a roof truss with internal loads and forces in each member are shown in Fig. Q3. Design the various members of the roof truss along with their end connection with bolt of property class 5.6. Also design the bearing plate at support for the reaction and anchor bolts for an uplift force of 15 kN. Also draw the Elevation of truss greater than half plan.

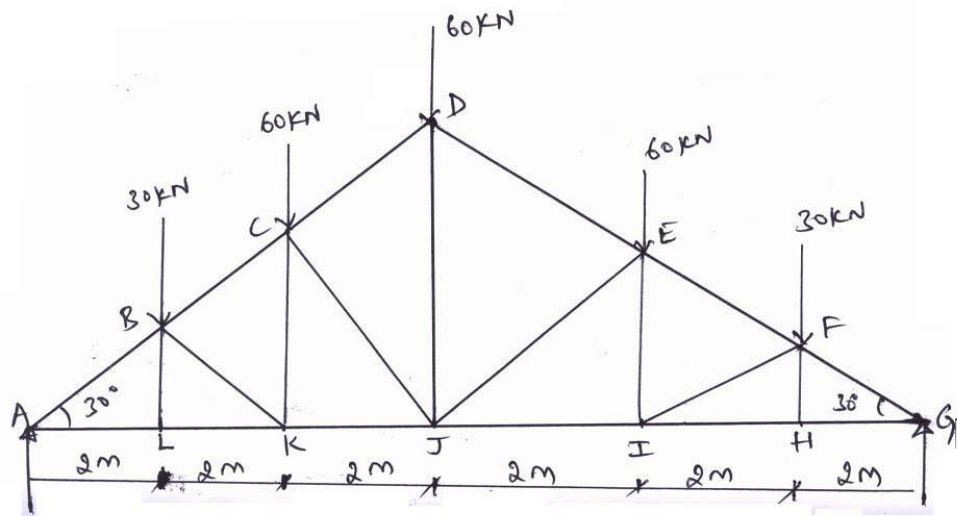


Fig. Q3

Members	Length (m)	Force (kN)	Nature of force
AB, GF	2.31	240	C
BC, FE	2.31	210	C
CD, ED	2.31	160.04	C
AL, GH	2.0	207.84	T
LK, HI	2.0	207.84	T
KJ, IJ	2.0	181.82	T
BL, FH	1.154	0	-
BK, FI	2.31	30	C
CK, EI	2.31	15	T
CJ, EJ	3.05	66	C
DJ	3.46	60	C

(50 Marks)

OR

- 4 Design a simply supported gantry girder to carry an electrically operated crane with the following data:

Span of Crane bridge = 25 m

Span of gantry girder = 8 m

Wheel base = 3.5 m

Crane capacity = 200 kN

Weight of crane bridge = 150 kN

Weight of trolley = 75 kN

Min Hook distance = 1.0 m

Weight of rail = 0.30 kN/m

Height of rail = 105 mm

Draw the sectional elevation.

(50 Marks)

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Seventh Semester B.E. Degree Examination, July/August 2022

Air Pollution and Control

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Air Pollution. Explain Primary and Secondary Air Pollutants. (10 Marks)
- b. Elaborate Photochemical Smog adding with Chemical Reactions. (10 Marks)

OR

- 2 a. Explain the effects of Air Pollution on Human Health and Vegetation. (10 Marks)
- b. Explain the effects of Air Pollution on Materials in detail. (10 Marks)

Module-2

- 3 a. List out the types of inversions and explain them. (10 Marks)
- b. Explain Plume behavior and type of Plume behavior with neat sketch. (10 Marks)

OR

- 4 a. Draw wind rose diagram for following data :

Wind directions	Duration of wind %		
	64 – 25 kmph	25 – 40 kmph	40 – 60 kmph
N	7.4	2.7	0.2
N N E	5.7	2.1	0.3
N E	2.4	0.9	0.6
E N E	1.2	0.4	0.2
E	0.8	0.2	0.0
E S E	0.3	0.1	0.0
S E	4.3	2.8	0.0
S S E	5.5	3.2	0.0
S	9.7	4.6	0.0
S S W	6.3	3.2	0.5
S W	3.6	1.8	0.3
W S W	1.0	0.5	0.1
W	0.4	0.1	0.0
W N W	0.2	0.1	0.0
N W	5.3	1.9	0.0
N N W	4.0	1.3	0.3

(10 Marks)

- b. A Thermal plant burns 5.45 tonnes with 4.2% sulphur per hour and discharge through a stack of effective height 7.5m. The average wind speed at top of stack 6m/s. Atmosphere is slightly to moderately stable. Find Ground Level Concentration (GLC) at 3km down wind and 0.4km cross wind distance. Take $\sigma_z = 170$ and $\sigma_y = 280$. (10 Marks)

Module-3

- 5 a. With neat sketches, explain the components of Sampling train. (10 Marks)
b. With the aid of neat sketch, elaborate working procedure of high volume Air samples. (10 Marks)

OR

- 6 a. Justify the necessity of Isokinetic sampling incase of stack sampling. (10 Marks)
b. Explain Preliminary considerations and stages of sampling. (10 Marks)

Module-4

- 7 a. Give the advantages and disadvantages of wet scrubber. (10 Marks)
b. Enumerate the design aspects of cyclone separators along with working principle. (10 Marks)

OR

- 8 a. Explain the construction and working of settling chambers. (10 Marks)
b. Explain with neat sketch, working of Fabric filters [BAG HOUSE FILTER]. (10 Marks)

Module-5

- 9 a. Write a note on Noise Pollution cases effects and control. (10 Marks)
b. Give the salient features of Environmental Protection Act 1986. (10 Marks)

OR

- 10 a. Explain briefly the emission of the gasoline driven vehicles and diesel driven vehicles. (10 Marks)
b. Describe the phenomena of “Green House effect” due to carbon dioxide. (10 Marks)

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Seventh Semester B.E. Degree Examination, July/August 2022

Pavement Materials & Construction

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. List and explain the desirable properties of road aggregates. State the tests conducted for each property. (10 Marks)
- b. With a neat sketch, explain impact test. (10 Marks)

OR

- 2 a. With the help of graph, explain Roth futch's method for gradation of aggregates. (10 Marks)
- b. Write the properties of bitumen. Any 4 explain. (04 Marks)
- c. List the differences of bitumen and tar. (06 Marks)

Module-2

- 3 a. Discuss the details of cutback and its types. (10 Marks)
- b. Explain the uses of bitumen emulsion. How are they prepared? (10 Marks)

OR

- 4 a. List and explain the mechanism of stripping. (10 Marks)
- b. Explain in detail, the static immersion test. (10 Marks)

Module-3

- 5 a. List and explain the requirements of bituminous mixes. (10 Marks)
- b. Explain how the bitumen content is determined. (Extraction test) (10 Marks)

OR

- 6 a. Explain the preparation and specifications of stability test by Marshall method. (08 Marks)
- b. The specific gravities and weight proportions for aggregates and bitumen are as under for the preparation of Marshall mix design. The volume and weight of one Marshall specimen was found to be 475 CC and 1100 gm. Assuming absorption of bitumen in aggregate is zero. Find V_v , V_b , VMA and VFB

Item	A - 1	A - 2	A - 3	A - 4	B
Wt (gm)	825	1200	325	150	100
Sp. gr.	2.63	2.51	2.46	2.43	1.05

(12 Marks)

Module-4

- 7 a. What is a power shovel? With a neat sketch, explain its operation and application. (10 Marks)
- b. List and explain the steps used in the preparation of subgrade. (10 Marks)

OR

- 8 a. What are the different types of compacting equipments used for pavement construction? Write an explanatory note on rollers in road construction. (10 Marks)
- b. What are the desirable properties subgrade soil? (04 Marks)
- c. List out the quality control tests for construction of highway subgrade. (06 Marks)

Module-5

- 9 a. Mention the specification of materials, construction steps and quality control tests for laying bituminous binder course. (12 Marks)
b. Explain the construction steps and specification for granular sub-base course. (08 Marks)

OR

- 10 a. What are the requirements of a good joint? Explain briefly with diagram of expansion joint. (10 Marks)
b. Briefly describe the quality control tests used in construction of concrete pavements. (10 Marks)

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18CV745

Seventh Semester B.E. Degree Examination, July/August 2022 Urban Transport Planning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the problems in the urban transportation in the present scenario. (10 Marks)
- b. Explain the features of BRTS. (06 Marks)
- c. Explain travel demand. (04 Marks)

OR

- 2 a. Explain with a flow chart on urban transportation system planning process. (10 Marks)
- b. Explain the types of transit systems. (10 Marks)

Module-2

- 3 a. What is sampling and explain the various types of samplings. (08 Marks)
- b. Explain the home interview surveys. (07 Marks)
- c. Explain the use of secondary sources. (05 Marks)

OR

- 4 a. Explain the factors considered for the selection of external cordon line for an urban transportation. (10 Marks)
- b. Write short notes on;
 - i) Commercial vehicle surveys.
 - ii) Taxi Surveys and Tag Survey. (10 Marks)

Module-3

- 5 a. Explain in detail the various factors governing trip generation. (10 Marks)
- b. What is multiple regression analysis and assumptions? (10 Marks)

OR

- 6 a. Define the following;
 - i) Trip Generation. (06 Marks)
 - ii) Home based and non home based trip. (05 Marks)
 - iii) Trip attraction.
- b. Mention the assumptions made in category analysis.
- c. Fit a linear equation relating trip rate and household size for the field survey data obtained given below. It was found that the house hold sizes are 1, 2, 3 and 4 and the trip ratio corresponding to household given in the table.

	Household size (x)			
	1	2	3	4
	1	2	4	6
Trips per day (y)	2	4	5	7
	2	3	3	4
Σy	5	9	12	17

(09 Marks)

Module-4

- 7 a. Explain in detail types of trip distribution methods. (10 Marks)
 b. Solve the future trips for the given data below by average growth factor method.

D O	A	B	C
A	30	50	100
B	50	10	150
C	100	150	10

The future trips generated in zones A, B and C are expected to be 180, 630 and 1560 respectively. (10 Marks)

OR

- 8 a. The total trips produced in and attracted to the three zones A, B and C of a survey area in the design year are tabulated as.

Zone	Trips produced	Trips attracted
A	1000	1500
B	1500	2000
C	2000	1000

It is known that the trips between two zones are inversely proportional to the second power of the travel time between zones, which is uniformly 10 minutes. If the trips interchange between B and C is known to be 300, calculate the trip interchange between zones A & B, A & C, B & A and C & B. (08 Marks)

- b. Explain in detail the Fratar method of finding trip distribution and mention the disadvantages. (06 Marks)
 c. What are opportunity model? Explain the types of opportunity model. (06 Marks)

Module-5

- 9 a. Describe the structure of a Lowry model, using a flow chart. (10 Marks)
 b. Explain the different types of Assignment Techniques for traffic assignment. (10 Marks)

OR

- 10 Write short notes on the following:

- Difficulties in Transports planning
- Diversion Curves
- Quick response transport planning technique
- Capacity restraint techniques.

(20 Marks)

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CBCS SCHEME

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Eighth Semester B.E. Degree Examination, July/August 2022 Design of Prestressed Concrete

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Distinguish between pre tensioning and post tensioning. (06 Marks)
- b. List the advantages of PSC over RCC. (04 Marks)
- c. Explain with sketch Hoyer's long line systems of pre-tensioning. (10 Marks)

OR

- 2 a. Explain concept of Thrust line. (06 Marks)
- b. A rectangular concrete beam of cross section 300mm deep and 200mm wide is prestressed by means of 15 wires of 5mm diameter located 65mm from the bottom of beam and 3 wires of 5mm diameter, 25mm from top. Assuming the prestress in steel as 840N/mm², calculate the stresses at the extreme fibres of mid span section. When the beam is supporting its own weight over a span of 6m. If a uniformly distributed live load of 6kN/m is imposed, evaluate the maximum working stress in concrete. (14 Marks)

Module-2

- 3 a. List different types of losses in post tensioning system. Explain any two. (06 Marks)
- b. A pretensioned beam, 200mm wide and 300mm deep is prestressed by 10 wires, of 7mm diameter initially stressed to 1200N/mm², with their centroids located 10mm from the soffit. Find the maximum stress in concrete immediately after transfer, allowing only for elastic shortening of concrete. If the concrete undergoes a further shortening due to creep and shrinkage while there is a relaxation of 5% of steel stress, estimate the final percentage loss of stress in the wires using the Indian Standard Code regulations and following data:
 $E_s = 210\text{kN/mm}^2$, $E_c = 5000\sqrt{f_{cu}}$, $f_{cu} = 42\text{ N/mm}^2$ creep co-efficient (ϕ) = 1.6,
 total residual shrinkage strain = 3×10^{-4} . (14 Marks)

OR

- 4 a. Explain:
 - i) Short term deflection
 - ii) Long term deflection
 - iii) Limiting deflection as per IS code. (06 Marks)
- b. A type-3 post tensioned pre stressed concrete beam of 10m span. The beam is post tensioned using three high tensile bars of 40mm diameter located @ an effective depth of 700mm. The effective cover from each of the vertical face of the beam is 60mm. the effective pre stressing force in each bar after all losses is 600kN. Given, cross section 450mm × 750mm, $\frac{x}{d} = 0.43$, $\frac{I_c}{b_d^3} = 0.081$, $f_p = 1035\text{N/mm}^2$, $E_s = 200\text{kN/mm}^2$, $E_c = 28\text{kN/mm}^2$, compute the width of cracks in the tension zone if the service load moment at mid span is 1040kN-m. (14 Marks)

Module-3

- 5 a. Explain failure modes of beam under flexure. (06 Marks)
- b. A post tensioned bridge girder with unbonded tendons is of box section of overall dimensions 1200mm wide by 1800mm deep, with wall thickness of 150mm. The high tensile steel has an area of 4000mm² and is located at an effective depth of 1600mm. The effective prestress in steel after all losses is 1000N/mm² and the effective span of girder is 24m. If $f_{ck} = 40\text{N/mm}^2$ and $f_p = 1600\text{N/mm}^2$, estimate the ultimate flexural strength of section. (14 Marks)

OR

- 6 Design a simply supported Type-1 pre stressed beam with total moment $M_T = 435\text{kN-m}$ including self weight moment of $M_{sw} = 55\text{kN-m}$. The height of the beam is restricted to 920mm. The pre stress at transfer $f_{p_o} = 1035\text{N/mm}^2$ and pre stress at service $f_{p_e} = 860\text{N/mm}^2$. The allowable compressive stresses are 12.5N/mm² at transfer and 11.0N/mm² at service. The pre-stressing tendon is 7 wire strand with nominal diameter of 12.8mm and nominal area of 99.3N/mm². (20 Marks)

Module-4

- 7 a. Explain different methods of improving shear resistance of PSC members. (05 Marks)
- b. Explain the mechanism of shear failure in PSC beams. (05 Marks)
- c. The support section of a PSC beam 150 × 300 is to resist a shear of 100kN. The pre stress at centroidal axis is 5N/mm², and $f_{ck} = 40\text{N/mm}^2$. The cover to tension reinforcement is 45mm. Check the section for shear and design suitable shear reinforcement using IS Code recommendation. (10 Marks)

OR

- 8 a. Differentiate between web shear and flexure shear cracks in PSC members with neat sketches. (06 Marks)
- b. A pre stressed I section has the following properties. Area = $55 \times 10^3\text{mm}^2$, $I = 189 \times 10^7\text{mm}^4$, statical moment about the centroid = $468 \times 10^4\text{mm}^3$, thickness of web = 50mm. It is prestressed horizontally by 24 wires of 5mm diameter and vertically by similar wires at 150mm centres. All the wires carry a tensile stress of 900N/mm². Calculate the principal stress at the centroid when shearing force of 80kN acts upon this section. (14 Marks)

Module-5

- 9 a. Write a note on anchorage zone stresses. (04 Marks)
- b. Explain end zone reinforcement. (06 Marks)
- c. The end block of a post-tensioned prestressed concrete beam, 300mm wide and 300mm deep, is subjected to a concentric anchorage force of 832.8kN by a Freyssinet anchorage of area 11720mm². Design and detail the anchorage reinforcement for the end block. (10 Marks)

OR

- 10 a. Explain with neat sketches the following pre-stressing systems:
i) Freyssinet's system ii) BBRV system. (10 Marks)
- b. The end block of a post-tensioned bridge girder is 600mm wide by 1200mm deep. Two cables, each comprising 97 high tensile wires of 7mm diameter, are anchored using square anchor plates of side length 410mm with their centres located at 600mm from the top and bottom edges of beam. The jacking force in each cable is 4500kN. Design a suitable anchorage zone reinforcement using Fe-415 grade HYSD bars conforming to IS:1343 code provisions. (10 Marks)

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Eighth Semester B.E. Degree Examination, July/August 2022 Advanced Foundation Engineering

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. IS : 6403 – 1981 is allowed.

Module-1

- 1 a. List and explain factors affecting the bearing capacity of soil. (10 Marks)
- b. A square footing located at a depth of 1.5m below the ground surface in cohesionless soil carries a column load of 1200kN. The soil is submerged having an effective unit weight of 11kN/m^3 and an angle of shearing resistance of 30° . Find the size of the following for factor of safety = 3 by Terzaghi's theory of general shear failure [$N_c = 37.2$, $N_q = 22.5$, $N_\delta = 19.7$]. (10 Marks)

OR

- 2 a. A footing, 2m square, rests on a soft clay soil with its base at a depth of 1.2m from ground surface. The clay stratum is 3.5m thick and is underlain by a sand stratum. The clay has the following properties. $W_L = 30\%$; $W_n = 40\%$; $G_s = 2.70$; $\phi = 0$; $C_u = 0.5\text{kg/cm}^2$ [50kN/m^2] [$N_c = 5.7$, $N_q = 1.0$, $N_\delta = 0$]. It is known that the clay stratum is normally consolidated. Compute the settlement that would result if this load intensity were allowed to act on the footing. Use Terzaghi's equation to determine safe bearing capacity [F.O.S = 3]. (12 Marks)
- b. Explain the steps involved in the proportioning of footings. (08 Marks)

Module-2

- 3 Design a combined footing having longitudinal beam joining two columns $400\text{mm} \times 400\text{mm}$ and $300\text{mm} \times 300\text{mm}$ carrying loads of 750kN and 300kN respectively. These columns are located at 4.5m apart c/c. The allowable soil pressure is 130kN/m^2 . (20 Marks)

OR

- 4 A building consists of 12 columns $400 \times 400\text{mm}$ sizes arranged in three rows of four each. The distance between the columns is 5m each. The load carried by four corner columns is 500kN each, that carried by exterior column is 550kN each and that carried by interior column is 900kN each. The allowable soil pressure is 50kN/m^2 . Design the raft foundation. (20 Marks)

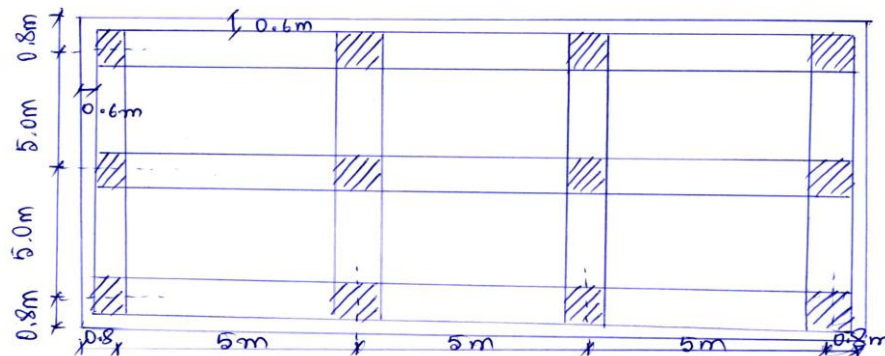


Fig.Q.4

Module-3

- 5 a. List and explain types of piles according to their composition, method of installation, and uses of piles. (08 Marks)
- b. A concrete pile of 0.45m diameter is driven through a system of layered cohesive soils. The length of the pile is 16m. Compute allowable pile load capacity with factor of safety = 2.5 using static formula. (12 Marks)

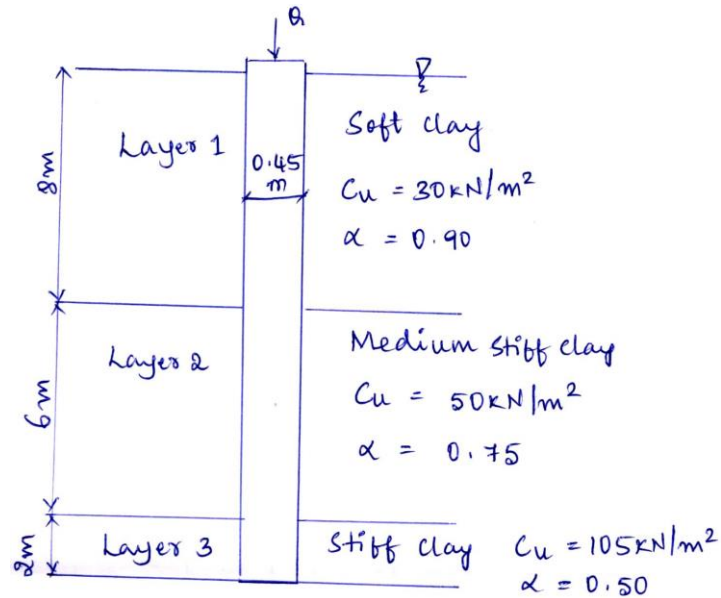
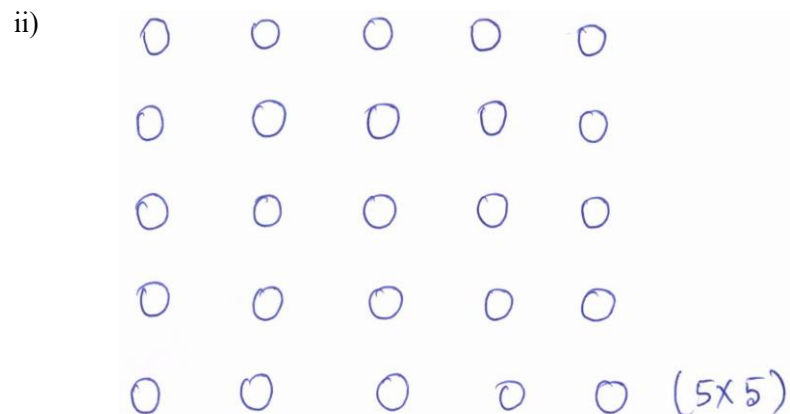
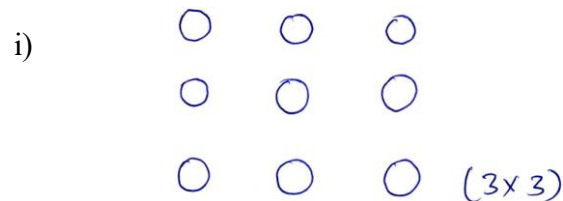


Fig.Q.5(b)

OR

- 6 a. Determine the group efficiency of piles using Feld's rule



(08 Marks)

- b. For the under reamed pile shown in Fig.Q.6(b), determine the safe load carrying capacity for tension load (F.O.S = 3). Pile diameter is 0.5m and bulb diameter is 1.25m. (12 Marks)

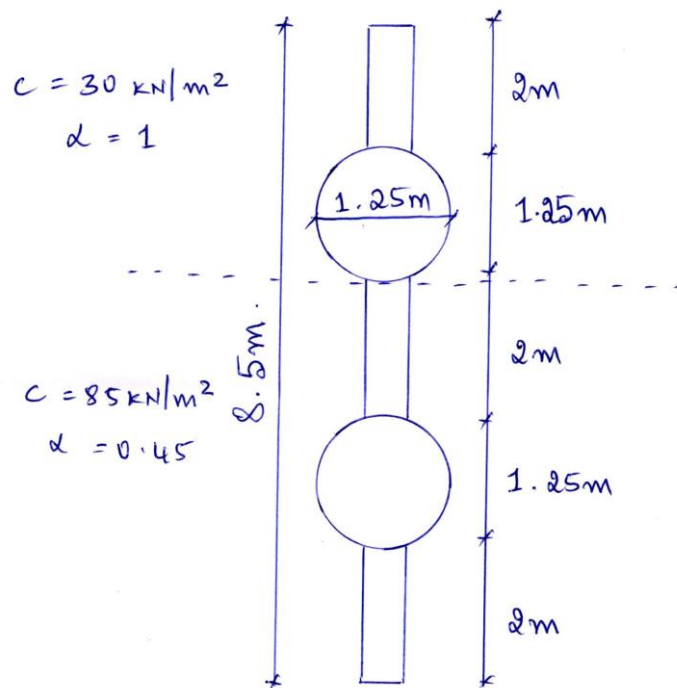


Fig.Q.6(b)

Module-4

- 7 a. Explain different shapes and characteristics of well foundation. (10 Marks)
b. List and explain forces acting on well foundation. (10 Marks)

OR

- 8 a. Explain sinking of wells. Write the methods to rectify tilt and shift of well. (10 Marks)
b. Explain the advantages and disadvantages of drilled piers and floating caissons. (10 Marks)

Module-5

- 9 a. Explain the types of machine foundations. (08 Marks)
b. Write the general criteria for design of machine foundation. (12 Marks)

OR

- 10 a. Explain vibration analysis of a machine foundation. (10 Marks)
b. Determine natural frequency and spring constant of machine foundation which has base area $3\text{m} \times 3\text{m}$ and weight of 20,000kg including weight of the machine. Take the value of coefficient of uniform compression as $4.4 \times 10^7 \text{ N/m}^3$. (10 Marks)

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Eighth Semester B.E. Degree Examination, July/August 2022

Rehabilitation and Retrofitting

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain how Freeze and Thaw on concrete causes deterioration of concrete structures. (10 Marks)
- b. Explain the following terms :

(i) Repair.

(ii) Maintenance.

(iii) Rehabilitation.

(iv) Retrofitting.

(v) Strengthening

(10 Marks)

OR

- 2 a. Write short notes on Durability of concrete and Acid attack. (08 Marks)
- b. Explain the evaluation of structural damages to the concrete structural elements due to earthquake. (12 Marks)

Module-2

- 3 a. Explain the step by step Damage Assessment Procedure. (08 Marks)
- b. What is destructive testing systems? Why load tests are usually carried? (04 Marks)
- c. Explain the penetration technique. (08 Marks)

OR

- 4 a. Write the objectives of rapid structured assessment. Explain the rapid assessment procedure. (12 Marks)
- b. List the common methods used in construction industry for semi-destructive testing systems. (08 Marks)

Module-3

- 5 a. Explain the effects due to climate and temperature. (08 Marks)
- b. Explain the cathode protection. (06 Marks)
- c. What are the effects due to corrosion? Write the Anode and Cathode reactions. (06 Marks)

OR

- 6 a. Explain the corrosion Inhibitors. (08 Marks)
- b. Explain the design and construction errors of concrete structures. (12 Marks)

Module-4

- 7 a. Discuss the importance of maintenance of structures. (08 Marks)
- b. Explain Beam Jacketing Technique, with sketches. (12 Marks)

OR

- 8 a. Explain externally bonding technique. (12 Marks)
- b. What is the process involved in near surface mounted technique. (08 Marks)

Module-5

- 9 a. What is carbon fibre reinforced polymer? (06 Marks)
- b. Explain the types of special mortars. (10 Marks)
- c. What are the properties of Epoxy resins? (04 Marks)

OR

- 10 a. What is Sisal Fibre and write its properties? (08 Marks)
- b. What is shoring and under Pinning? Explain its types. (12 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. 42+8 = 50, will be treated as malpractice.